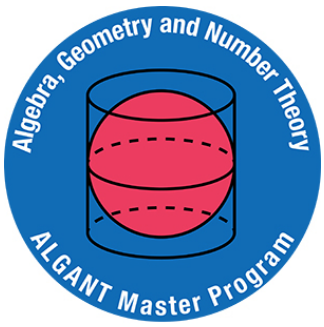


Università degli studi di Padova
Dipartimento di Matematica "Tullio Levi-Civita"

Université de Bordeaux
Institut de Mathématiques de Bordeaux

ZETA FUNCTIONS OF CODES IN THE HAMMING AND RANK-METRICS

Disha Atukuri

Supervisor:

Dr. Elena Berardini

Université de Bordeaux







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Abstract

Motivated by the properties of algebraic curves and the linear codes constructed from them (called algebraic geometry codes), Duursma introduced and studied, in a series of articles, analogues of the zeta function and its L-polynomial for linear codes in the Hamming metric. This project explores the zeta functions of codes in the Hamming and rank-metrics using properties analogous to or equivalent to those known for the zeta functions of algebraic curves.

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Introduction

In today's world of AI and computers, the security of digital communication remains one of the biggest problems of our time. Everything from banking and healthcare to government services and social interactions runs over digital networks. This is why a breach in communication security can have massive real-world consequences. Prime number-based cryptography, for example, RSA, underpins much of our current digital communication security. In simple words, a better understanding of prime numbers could change the security landscape of cryptography. But this is not the first time that prime numbers have been the center of attention, especially for mathematicians.

In 1859, Bernhard Riemann introduced the Riemann zeta function, which would be used as a tool for studying prime numbers. Then came the Riemann hypothesis, a conjecture about the distribution of the zeros of the Riemann zeta function. It (arguably) remains the most popular and important unsolved problem in Number Theory for almost two centuries.

It is not unusual in mathematics for seemingly distinct areas to converge and interact in meaningful ways. Ideas that arise in one field may be used to understand mathematical objects in another field. This thesis report explores one such bridge, inspired by the Riemann zeta function. It aims to understand the theory of zeta functions of linear codes, which lie at the intersection of analytic number theory, algebraic geometry, and coding theory.

In a beautiful mathematical coincidence, codes, like prime numbers, also underpin digital communication. Error-correcting codes, i.e., codes that transmit information with less error or information loss, ensure that messages pass reliably over noisy channels, and prime number-based cryptographic systems aim to keep those messages secure.

The zeta function of a linear code, first defined by Duursma in [5], encodes information about the weight distribution of a code. It helps us understand the global structure of a code, i.e., how many codewords of a certain weight it may have. Duursma [2] showed that codes which satisfy the Riemann hypothesis analogue for coding theory, have extremal or more optimal properties, such as achieving the Mallows-Sloane-Duursma bound (Theorem 1.2).

The definition, structure, and properties of the zeta function of a linear code have been developed analogously to the zeta function of an algebraic curve over a finite field [5]. The Riemann hypothesis analogue for algebraic curves over finite fields was famously solved by A. Weil in the 1940s.

The objective of this thesis is to understand the structure and theory of the zeta functions of linear codes in the Hamming and rank-metrics, and consequently the Riemann hypothesis analogue for coding theory.

In Chapter 1, we introduce all necessary notations and results related to linear codes in the Hamming and rank-metrics, with a focus on their weight enumerators. We understand the theory of zeta functions of algebraic curves over finite fields, and the Riemann hypothesis analogue in depth. In Chapter 2, we introduce the zeta function of Hamming metric codes. We will give three definitions (Definition 2.1, 2.3, and 2.6) for it, all due to Duursma, and then calculate its degree, prove its existence and uniqueness (Theorem 2.2),

obtain its functional equation (2.4), and define some of its properties (Subsection 2.4). Following this, we highlight the importance of the negative sum of the reciprocal zeros of the Duursma zeta function, and obtain a Riemann hypothesis analogue for codes. We focus in particular on self-dual Hamming metric codes (Subsection 2.5.2) since the structure of the functional equation of their zeta function bears closer analogies to that of algebraic curves over finite fields. We will end the section with a conjectured result by Duursma about the Riemann hypothesis analogue for extremal self-dual linear codes (Theorem 1.2).

In Chapter 3, we introduce the zeta functions of rank-metric codes. The results and notations of this Chapter closely follow [3]. Analogous to the Hamming case, we give three definitions of its zeta function (Theorem 2.1 and Definition 3.4), all equivalent, and we obtain its functional equation (Theorem 3.3). Again, analogous to the Hamming case, we highlight the importance of the negative sum of reciprocal zeros of its zeta function. But unlike the Hamming metric case, a clean analogue of the Riemann hypothesis does not exist for rank-metric codes. Throughout this report, we will explicitly highlight the analogies between the zeta functions of linear codes in the Hamming and rank-metrics and the zeta function of algebraic curves.

Chapter 1

Preliminaries and notations

In the first section of this chapter, we recall the Riemann zeta function and the Riemann hypothesis. Following that, we define a linear code and its properties. We particularly focus on linear codes in the Hamming metric and in the rank- metric. We want to study the relation of their zeta functions with their weight enumerator polynomials, both of which are arithmetic invariants of linear codes. The weight enumerator polynomial is fundamental for obtaining information about the structure of linear codes, specifically their weight distribution, i.e., the number of codewords of a given weight. Thus, it is studied in great detail. In the following chapter, we define the zeta function of a linear code. It encodes the same information as the weight enumerator polynomial.

In the third section, we define an algebraic curve, its divisors and zeta function, and give an analogue of the Riemann Hypothesis for algebraic curves over finite fields. The theory of the zeta function of a linear code has been developed analogously to the theory of the zeta function of an algebraic curve.

1.1 The Riemann zeta function

In this section, we will briefly recall the Riemann zeta function and the Riemann hypothesis, for motivation. We will observe later, as we define the Hasse-Weil zeta function of algebraic curves and the Duursma zeta function of linear codes, the similarities in the structure of their zeta functions and their corresponding Riemann hypothesis analogues.

The Riemann zeta function $\zeta(s)$ is defined for complex variables s by the following absolutely convergent infinite series:

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

for values of s with $\mathrm{Re}(s) > 1$.
In the strip $0 < \mathrm{Re}(s) < 1$, outside the region of convergence of this series, the zeta function satisfies the functional equation:

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s)\zeta(1-s).$$

This equation holds for all complex s except $s = 0, 1$, where $\zeta(s)$ has simple poles.
If we let $\xi(s) = \pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s)$, then we can rewrite this as

$$\xi(1-s) = \xi(s).$$

If s is a negative even integer, then $\zeta(s) = 0$, because the factor $\sin(\pi s/2)$ vanishes. These are the *trivial zeros* of the zeta function (namely $s = -2, -4, -6, \ldots$). The Riemann hypothesis focuses on the non-trivial zeros and states that:

"The real part of any nontrivial zero of the Riemann zeta function is $\frac{1}{2}$*."*

The line $\mathrm{Re}(s) = 1/2$ is called the *critical line*.

1.2 Linear codes

In this section, we will recall the definitions of Hamming and rank-metric codes, and their weight enumerators, required to define their zeta functions. The notations and results of this section closely follow those of [1].

A *linear code* C of length N over the alphabet $\mathbb{F}_q$ is defined as a linear subspace of $\mathbb{F}_q^N$, where $q = p^r$ (p prime). We will define two types of metrics on linear codes: the Hamming metric and the rank-metric.

1.2.1 Hamming metric

A linear code C in the Hamming metric, of length n over the alphabet $\mathbb{F}_q$ is defined as a linear subspace of $\mathbb{F}_q^n$. An element c of C is a vector of the form $(c_1, \ldots, c_n) \in \mathbb{F}_q^n$, known as a *codeword*. The *length n of a code* corresponds to the length of a codeword, and the *dimension* k of a code corresponds to the dimension of C as an $\mathbb{F}_q$ vector space. If $x \in C$, $y \in C$, then the *Hamming distance* $\mathrm{d}(x, y)$ of x and y is defined by

$$d(x, y) := |\{i | 1 \leq i \leq n, x_i \neq y_i\}|.$$

The weight $w(x)$ is defined as $w(x) := d(x, 0)$. The *minimum distance* of a code is defined as

$$d := \min\{d(x, y) | x \in C, y \in C, x \neq y\},$$

the minimum of all the Hamming distances. The *minimum weight* of C is defined as

$$\min\{w(x) | x \in C, x \neq 0\}.$$

Remark 1.1. For a linear code, the minimum distance is equal to the minimum weight.

The *dual* $C^\perp$ of C is defined by

$$C^\perp := \{y \in \mathbb{F}_q^n | \forall x \in C, \langle x, y \rangle = \sum_{i=1}^{n} x_i y_i = 0\}.$$

The dual code is also a linear code with length n and dimension $n - k$. If $C = C^\perp$, then C is called a *self-dual code*. The triple $[n, k, d]_q$ is referred to as the *parameters of the code* C. The *genus* of an $[n, k, d]_q$−code C is defined by $\gamma(C) = n + 1 - k - d$. This measures how far the code is from being an MDS (maximum distance separable) code.

A code is said to be *maximum distance separable* if it achieves the *Singleton bound*, i.e., $k + d \leq n + 1$ with equality. The singleton bound also gives an upper bound on the size of a code, i.e., $|C| = q^k \leq q^{n+1-d}$. If C is a self-dual linear code, then $k = n - k$, which implies that $k = n/2$, and its genus satisfies

$\gamma = n + 1 - n/2 - d = n/2 + 1 - d$. A combination of the singleton bound for a code, $d + k \leq n + 1$ and its dual $d^{\perp} + (n - k) \leq n + 1$ gives

$$d + d^{\perp} \leq n + 2. \tag{1.1}$$

Definition 1.1. The *Hamming weight enumerator polynomial* of a linear code of length n and minimum distance d over $\mathbb{F}_q$ is given by

$$A_C(x, y) = x^n + \sum_{i=d}^{n} A_i x^{n-i} y^i$$

where $A_i = |\{c \in C | \, \mathrm{wt}(c) = i\}|$.

Definition 1.2. If $A_C(x, y) = A_{C^{\perp}}(x, y)$ then C is called a *formally self-dual linear code*.

Definition 1.3. Two codes are *formally equivalent* if they have the same Hamming weight enumerator polynomial.

Moving forward, we will refer to the Hamming weight enumerator polynomial simply as the weight enumerator. The *Mac Williams identity* relates the weight enumerator of C to the weight enumerator of its dual $C^{\perp}$ by

$$A_{C^{\perp}}(x, y) = |C|^{-1} A_C(x + (q - 1)y, x - y). \tag{1.2}$$

Definition 1.4. An *MDS weight enumerator* is a weight enumerator corresponding to parameters achieving the last bound given by (1.1). It satisfies $(d-1) + (d^{\perp} - 1) = n$ independent linear conditions and is uniquely determined by its length n and minimum distance d. It is denoted by $M_{n,d}$. Its coefficient at $x^{n-w} y^w$ is given by

$$\binom{n}{w}(q - 1) \sum_{j=0}^{w-d} (-1)^j \binom{w - 1}{j} q^{w-d-j}.$$

Generally, an MDS weight enumerator may have negative coefficients and may not be realized by an actual code. But we still find it interesting because they are closed under puncturing and shortening and the Mac William transform. We will go into detail on puncturing and shortening in the following sections. They can also be used to provide a basis on which we can express an arbitrary weight enumerator of length n in the following way:

$$A_C(x, y) = a_0 M_{n,d} + a_1 M_{n,d+1} + \cdots + a_r M_{n,d+r}. \tag{1.3}$$

The weight enumerators $M_{n,d}$ and $M_{n,n+2-d}$ are in duality under the Mac Williams transform.

Definition 1.5. Let $\mathrm{supp}(C) = \{\mathrm{wt}(c) : c \in C\}$ be the set of Hamming weights occurring in C. For a code C, if $b > 1$ is an integer and $\mathrm{supp}(C) \subset b\mathbb{Z}$, i.e., $b|i$ for all $c \in C$ with $\mathrm{wt}(c) = i$, then C is called *b-divisible*.

1.2.2 Rank-metric

Let m, n be two positive integers with $n \leq m$ and $q = p^r$ (p prime). We denote the $m \times n$ matrices with entries in $\mathbb{F}_q$ by $\mathbb{F}_q^{m \times n}$. The right null space of any $X \in \mathbb{F}_q^{m \times n}$ is denoted by $\ker(X)$ or $X^{\perp}$, and is given by $X^{\perp} := \{y \in \mathbb{F}_q^n : Xy^T = 0\}$. A linear code C in the rank-metric is defined as an $\mathbb{F}_q$-linear subspace of $\mathbb{F}_q^{m \times n}$. An element c of C is an $m \times n$ matrix with entries in $\mathbb{F}_q$, known as a *codeword*.

The *rank distance* between matrices $X, Y \in \mathbb{F}_q^{m \times n}$ is $d(X, Y) := \mathrm{rk}(X - Y)$. For $|C| \geq 2$, the *minimum rank distance* of C is the integer defined by $d = d(C) := \min\{d(X, Y) : X, Y \in C, X \neq Y\}$.

Definition 1.6. The *weight distribution* of C is the integer vector $W(C) = (W_t(C) : 0 \le t \le n)$, where, for all $t \in \{0, \dots, n\}$,

$$W_t(C) := |\{X \in C : \mathrm{rk}(X) = t\}|.$$

The *rank-metric weight enumerator* of C is the bivariate polynomial

$$W_C(x, y) = \sum_{t=0}^{n} W_t(C) x^{n-t} y^t. \tag{1.4}$$

The *dual code* of C is the $\mathbb{F}_q$-linear code

$$C^\perp := \{Y \in \mathbb{F}_q^{m \times n} : \mathrm{Tr}(XY^t) = 0 \forall X \in C\}.$$

The map $(X, Y) \mapsto \mathrm{Tr}(XY^t)$ defines an inner product on the space $\mathbb{F}_q^{m \times n}$, so we have $\dim(C^\perp) = mn - \dim(C)$ and $C^{\perp\perp} = C$.

The rank-metric analogue of the singleton bound is given by $|C| \le q^{m(n-d+1)}$ for any C of minimum distance d. Rank-metric codes that meet this bound are called *maximum-rank-distance* (MRD) codes. The weight enumerator of an MRD code in $\mathbb{F}_q^{m \times n}$ of minimum distance d was shown in [16] to be uniquely given by

$$M_{n,d} := x^n + \sum_{t=d}^{n} \sum_{i=d}^{t} (-1)^{t-i} q^{\binom{t-i}{2}} \binom{n}{t} \binom{t}{i} (q^{m(i-d+1)} - 1) x^{n-t} y^t, \tag{1.5}$$

where the coefficients of $x^{n-t}y^t$ count the number of matrices in the MRD code of rank t. For a fixed n, the $M_{n,d}$ are linearly independent over $\mathbb{Q}$, ensuring the expansion (1.6) is unique. A code C whose dual code has minimum distance at least 2 has a weight enumerator that can be expressed as a $\mathbb{Q}-$linear combination of the MRD weight enumerators. That is, there exists $p_0, \dots, p_{n-d} \in \mathbb{Q}$ satisfying

$$W_C(x, y) = p_0 M_{n,d}(x, y) + \cdots + p_{n-d} M_{n,n}(x, y), \tag{1.6}$$

analogous to (1.3).

1.3 Virtual weight enumerators

In this section, everything defined and stated will be with respect to linear codes C in the Hamming metric. The Hamming weight enumerator polynomial we defined in Section 1.2.1 for a given code C can be generalized as follows.

Definition 1.7. A homogeneous polynomial $F(x, y) = x^n + \sum_{i=1}^{n} A_i x^{n-i} y^i$ of degree n with complex coefficients is called a *virtual weight enumerator* with *support* $\mathrm{supp}(F) = \{0\} \cup \{i | A_i \neq 0\}$. If $F(x, y) = x^n + \sum_{i=d}^{n} A_i x^{n-i} y^i$ with $A_d \neq 0$, then we call n the *length* of F and d the *minimum distance* of F. Such an F of even degree satisfying the following equation

$$F(x, y) = F\left(\frac{x + (q-1)y}{\sqrt{q}}, \frac{x - y}{\sqrt{q}}\right), \tag{1.7}$$

is called a *virtually self-dual weight enumerator* over $\mathbb{F}_q$ having genus $\gamma(F) = n/2 + 1 - d$.

Definition 1.8. If $b > 1$ is an integer and $\mathrm{supp}(F) \subset b\mathbb{Z}$, then the virtual weight enumerator F is called

b-divisible.

Let C be a b-divisible code. If C and $C^{\perp}$ are both binary and contain the all-ones codeword, then C is said to be *Type 2 divisible*. We say that C is *Type 1 divisible* if C is not of Type 2. Type 1 codes need not be binary. If C is *Type 1 divisible*, then $d + bd^{\perp} \leq n + b(b+1)$. If C is *Type 2* divisible, then $2d + bd^{\perp} \leq n + b(b+2)$. If C is formally self-dual b-divsible, then

$$d \leq \begin{cases} \left[\frac{n}{b+1}\right] + b, & \text{Type 1,} \\ \left[\frac{n}{b+2}\right] + b, & \text{Type 2.} \end{cases}$$

Let C be a formally self-dual b-divisible $[n, k, d]_q$-code. We say that C is of *Type I (singly even or formally self-dual even)* if $q = b = 2$, and n is even, *Type II (or doubly even)* if $q = 2$, $b = 4$, and $8|n$, *Type III* if $q = b = 3$, and $4|n$, *Type IV* if $q = 4$, $b = 2$, and n is even. The following theorem gives an analogue of this classification using the virtual weight enumerator.

Theorem 1.1 (Gleason-Pierce-Assmus-Mattson [10])**.** *Let F be a b-divisible virtually self-dual weight enumerator over $\mathbb{F}_q$. Then either*

1. *q=b=2*
2. *q=2, b=4,*
3. *q=b=3,*
4. *q=4, b=2,*
5. *q is arbitrary, b=2, and $F(x, y) = (x^2 + (q-1)y^2)^{n/2}$.*

Proof. A detailed proof for this theorem can be found in the survey paper by Sloane, [6]. □

Definition 1.9. If F is a b-divisible virtually self-dual weight enumerator over $\mathbb{F}_q$, then F is called *Type I* if $q = b = 2$ and $2|n$, *Type II* if $q = 2$, $b = 4$, and $8|n$, *Type III* if $q = b = 3$ and $4|n$, and *Type IV* if $q = 4$, $b = 2$, and $2|n$.

The following theorem gives the upper bounds on the minimum distance d for virtually self-dual weight enumerators of a given type.

Theorem 1.2 (Sloane-Mallows-Duursma [11], [12])**.** *If F is a b-divisible virtually self-dual weight enumerator with length n and minimum distance d, then*

$$d \leq \begin{cases} b\left[\frac{n}{b(b+1)}\right] + b, & \text{if } F \text{ is Type 1,} \\ b\left[\frac{n}{b(b+2)}\right] + b, & \text{if } F \text{ is Type 2.} \end{cases}$$

In particular,

$$d \leq \begin{cases} 2\left[\frac{n}{8}\right] + 2, & \text{if } F \text{ is Type I,} \\ 4\left[\frac{n}{24}\right] + 4, & \text{if } F \text{ is Type II,} \\ 3\left[\frac{n}{12}\right] + 3, & \text{if } F \text{ is Type III,} \\ 2\left[\frac{n}{6}\right] + 2, & \text{if } F \text{ is Type IV.} \end{cases}$$

For a detailed proof, one can refer to Appendix 7.4 of [4].

Definition 1.10. A virtually self-dual weight enumerator F is defined as *extremal* if any one of the bounds in Theorem 1.2 holds with equality.

It is important to note that we are talking about two different notions while discussing bounds on the minimum distance. The first is the notion of *extremal divisible*, and the second is the notion of *extremal self-dual*. In this report, we will focus on *extremal self-dual linear codes*, as defined in 1.10.

Remark 1.2. A virtual weight enumerator F is formally identified with an object we call a *virtual code* C, defined by the condition that its weight enumerator is $A_C = F$. If F happens to be the weight enumerator of an actual code C', then C and C' are formally equivalent (i.e., they have the same weight enumerator). In other words, a virtual code is only well-defined up to formal equivalence. If C_1 and C_2 are virtual codes, we define $C_1 + C_2$ to be the virtual code associated to the virtual weight enumerator $A_{C_1}(x, y) + A_{C_2}(x, y)$.

1.4 Algebraic curves

In this section, we will define an algebraic curve (projective, smooth, absolutely irreducible) of genus g over a finite field $\mathbb{F}_q$, its divisor theory, its zeta function, and its Riemann hypothesis analogue. The notations and results in this section closely follow those of [8] and [17].

Let k be an algebraically closed field. The n-dimensional *affine space* over a field k, denoted by $\mathbb{A}^n(k)$, is defined as the set of n-tuples $(x_1, \ldots, x_n)$ with coordinates from k.

The *projective n-space* over a field k, denoted by $\mathbb{P}^n(k)$, is defined as the set of equivalence classes of points of $\mathbb{A}^{n+1}(k)$ under the equivalence relation given by $(x_0, \ldots, x_n) \sim (\lambda x_0, \ldots, \lambda x_n)$ for every $\lambda \in k^*$. We denote the points in $\mathbb{P}^n(k)$ as $[x_0 : \ldots : x_n]$.

In the space $\mathbb{A}^n(k)$, we introduce a topology, the so-called *Zariski topology*. The *closed sets (algebraic sets)* are the zero sets of ideals J of $k[x_1, \ldots, x_n]$, i.e.

$$Z(J) := \{(x_1, \ldots, x_n) \in \mathbb{A}^n(k) | f(x_1, \ldots, x_n) = 0 \text{ for all } f \in J\}.$$

We always assume that J is *radical*. Recall that an ideal I is said to be *radical* if $\forall\, n \in \mathbb{N}, f^n \in I \implies f \in I$.

A closed subset C is called *irreducible* if C cannot be written as the union of two proper closed subsets of C. The set $Z(J)$ is irreducible if J is a prime ideal. An open set is the complement of a closed set.

In projective spaces, to define the notion of closed sets, we use homogeneous ideals. An ideal $I \subset k[x_0, \ldots, x_n]$ is said to be *homogeneous* if it is generated by homogeneous polynomials, i.e. polynomials f of the form

$$f(\lambda x_0, \ldots, \lambda x_n) = \lambda^{\deg f} f(x_0, \ldots, x_n) \text{for every} \lambda \in k.$$

The closed sets of Zariski topology in projective spaces are the zero sets of the homogeneous ideals.

A *projective variety* $\mathcal{X}$ is defined as the zero set of a homogeneous prime ideal p in $k[x_0, \ldots, x_n]$. The ring $k[x_1, ..., x_n]/p$ is called the *coordinate ring* $k[\mathcal{X}]$ of the variety $\mathcal{X}$. The coordinate ring is an integral domain since p is a prime ideal. Let $p \subseteq k[x_0, \ldots, x_n]$ be the homogeneous prime ideal defining $\mathcal{X}$, and let $R(\mathcal{X})$ be the subring of $k(x_0, \ldots, x_n)$ consisting of fractions f/g where f and g are homogeneous polynomials of the same degree and $g \notin p$. The *function field* $k(\mathcal{X})$ is the field of fractions of $R(\mathcal{X})$, i.e., $k(\mathcal{X}) = \text{Frac}(R(\mathcal{X})) = \{ f/g \mid f, g \text{ homogeneous of the same degree, } g \notin p \}$. A *rational function field*

over k is the function field $k(x)$ of the projective line $\mathbb{P}^1_k$ whose elements are rational functions $f(x)/g(x)$ with $f(x), g(x) \in k[x]$ and $g(x) \neq 0$, and it has genus 0.

Consider the standard affine chart $U_0 = \{[x_0 : \cdots : x_n] \mid x_0 \neq 0\} \cong \mathbb{A}^n$. The points of $\mathcal{X} \cap U_0$ are called the *affine points* of $\mathcal{X}$ (in this chart), and the complementary set $\mathcal{X} \setminus U_0 = \mathcal{X} \cap \{x_0 = 0\}$ is called the *points at infinity* of $\mathcal{X}$. By setting $x_0 = 1$, points in U_0 are identified with $(1, x_1, \ldots, x_n) \mapsto (x_1, \ldots, x_n) \in \mathbb{A}^n$, giving an explicit identification of $\mathcal{X} \cap U_0$ with an affine variety.

The *dimension* of the projective variety $\mathcal{X}$ is the transcendence degree of $k(\mathcal{X})$ over k. If this dimension is 1, then $\mathcal{X}$ is called a *projective curve*.

The *local ring* $\mathcal{O}_P$ of a point $P \in \mathcal{X}$ is the set of rational functions on $\mathcal{X}$ that are regular at P. The ring $\mathcal{O}_P$ has a unique maximal ideal m_P, consisting of functions in $\mathcal{O}_P$ that vanish at P. For an affine neighborhood U of P and homogeneous polynomials f, g of the same degree with $g(P) \neq 0$, the quotient $\phi = f/g$, defined on U, is regular at P.

Suppose we have a projective curve $\mathcal{X} \subseteq \mathbb{P}^n(k)$ defined by a system of $n-1$ homogeneous polynomials: $F_1(x_0, \ldots, x_n) = 0, \quad F_2(x_0, \ldots, x_n) = 0, \quad \ldots, \quad F_{n-1}(x_0, \ldots, x_n) = 0,$, with the condition that the polynomials are independent. A point $P \in \mathcal{X}$ is *smooth* if the Jacobian matrix

$$\left(\frac{\partial F_i}{\partial x_j}(P)\right)_{\substack{1 \leq i \leq n-1 \\ 0 \leq j \leq n}}$$

has rank $n-1$. A projective curve $\mathcal{X}$ is called a *smooth projective curve* if every point $P \in \mathcal{X}$ is a smooth point.

For a smooth point $P \in \mathcal{X}$, the local ring $\mathcal{O}_P$ is a discrete valuation ring. Its unique maximal ideal m_P is called a *place* of the function field $k(\mathcal{X})/\mathbb{F}_q$. The residue field $\mathcal{O}_P/m_P$ is a finite extension of $\mathbb{F}_q$. Its degree $[\mathcal{O}_P/m_P : \mathbb{F}_q]$ is called the *degree* of P and is denoted by $\deg P$. The *cotangent space* at P is the k-vector space m_P/m_P^2. Its dual, the *tangent space* $T_P\mathcal{X}$, consists of all k-linear maps $m_P/m_P^2 \to k$. A *local parameter* (uniformizer) at P is any element $t \in m_P$ whose image in m_P/m_P^2 is non-zero. Equivalently, t generates m_P and its class dt (in m_P/m_P^2) spans the cotangent space.

Now that we have defined the notion of a smooth projective curve, we define its genus g. For this report, the following method to calculate the genus g of a smooth projective curve of degree d in $\mathbb{P}^2$, given by the following formula, suffices:

$$g = \frac{1}{2}(d-1)(d-2).$$

This is known as the Plücker formula.

From now on, let k be the algebraic closure of $\mathbb{F}_q$. We particularly want to work with *smooth projective curves* $\mathcal{X}$ of genus g over a finite field $\mathbb{F}_q$. This means that $\mathcal{X}$ is defined by homogeneous polynomials with coefficients in $\mathbb{F}_q$. The points of $\mathcal{X}$ are equivalence classes of tuples in k^{n+1} (up to scaling) satisfying these equations. A point $P \in \mathcal{X}$ is called an $\mathbb{F}_q$*-rational point* (or simply a *rational point*) if it can be represented by homogeneous coordinates $[x_0 : \ldots : x_n]$ with all $x_i \in \mathbb{F}_q$.

For the divisor theory and zeta function that follow, we also require $\mathcal{X}$ to be absolutely irreducible. A projective curve is *absolutely irreducible* if its defining homogeneous polynomial remains irreducible over any finite extension of $\mathbb{F}_q$. Equivalently, when viewed over the algebraic closure k of $\mathbb{F}_q$, the curve does not decompose into a union of two curves. Moving forward, $\mathcal{X}$ will denote an *algebraic curve* (smooth, projective, absolutely irreducible) curve of genus g over $\mathbb{F}_q$.

Example 1.1. Let's consider an elliptic curve defined by $F(x, y, z) = y^2z + yz^2 - x^3 - x^2z = 0$ in $\mathbb{P}^2$, over

$\mathbb{F}_2$. In the affine chart $z = 1$, this becomes $y^2 + y - x^3 - x^2 = 0$, and we can see that $F_y = 2y + 1 = 1 \neq 0$, so $\mathcal{X}$ is smooth. The polynomial F is absolutely irreducible over $\mathbb{F}_2$ (it remains irreducible over $k = \mathbb{F}_2$). And since $\mathcal{X}$ is defined by a degree 3 polynomial, we can use Plücker formula to calculate its genus to be 1. This curve consists of five $\mathbb{F}_2$ rational points, out of which there are four affine points: $(0, 0)$, $(0, 1)$, $(1, 0)$, and $(1, 1)$, and one point at infinity: $[0 : 1 : 0]$.

1.4.1 Divisors

A *divisor* D of $\mathcal{X}$ is defined as a formal sum $D = \sum_{P \in \mathbb{P}_\mathcal{X}} n_P P$, where $n_P \in \mathbb{Z}$ and $n_P = 0$ for all but finitely many points P. $\mathrm{Div}(\mathcal{X})$ is the *additive group of divisors* with formal addition (the free abelian group on $\mathcal{X}$). The *support* of D, denoted by $\mathrm{supp}(D)$, is the set of P with $n_P \neq 0$. The *degree* $\deg(D)$ of a divisor is $\deg(D) = \sum_{P \in \mathbb{P}_\mathcal{X}} n_P \cdot \deg P$, where $\deg P$ is defined as the degree of the *residue field extension* $k_P/\mathbb{F}_q$, with $k_P = \mathcal{O}_P/m_P$ the residue field at P. The *order* of D at P is defined as $v_P(D) = n_P$.

For a non-zero rational function $f \in k(\mathcal{X})^*$, we define its *principal divisor* as $(f) := \sum_{P \in \mathbb{P}_\mathcal{X}} v_P(f) \cdot P$, where $v_P(f)$ is the *order of vanishing (or pole)* of f at the point P. If $v_P(f) > 0$, P is called a zero of f, and if $v_P(f) < 0$, P is a pole of f. The degree of a divisor of a non-zero rational function is zero. The set of all principal divisors, $\mathrm{Princ}(\mathcal{X}) := \{(f) \mid f \in k(\mathcal{X})^*\}$, is a subgroup of $\mathrm{Div}(\mathcal{X})$.

Two divisors D and D' are called *equivalent*, written $D \sim D'$, if $D - D' = (f)$ for some non-zero rational function $f \in k(\mathcal{X})^*$. The set of equivalence classes $\mathrm{Cl}(\mathcal{X}) := \mathrm{Div}(\mathcal{X})/\mathrm{Princ}(\mathcal{X})$ is called the *divisor class group* of $\mathcal{X}$. For a divisor D, we denote its class by $[D] \in \mathrm{Cl}(\mathcal{X})$. The degree map $\deg : \mathrm{Div}(\mathcal{X}) \to \mathbb{Z}$ descends to $\mathrm{Cl}(\mathcal{X})$ because principal divisors have degree 0. Hence, we can speak of the degree $\deg[C]$ of a class $[C]$. The subgroup of classes of degree 0 is denoted by $\mathrm{Cl}^0(\mathcal{X}) := \{[C] \in \mathrm{Cl}(\mathcal{X}) \mid \deg[C] = 0\}$. It is a finite group. Its order is called the *class number* of $\mathcal{X}$ and is denoted by $h := \#\,\mathrm{Cl}^0(\mathcal{X})$.

A divisor $D = \sum_{P \in \mathbb{P}_\mathcal{X}} n_P P$ is called *effective* (or *positive*) if $n_P \geq 0$ for all points $P \in \mathcal{X}$.

A divisor of the form P (i.e., $1 \cdot P$) for a single point $P \in \mathbb{P}_\mathcal{X}$ is called a *prime divisor*. In the language of algebraic geometry, these correspond to the *closed points* of $\mathcal{X}$. For each prime divisor P, the integer $\deg P$ is the degree of the point P as defined earlier. Let ∂ be the greatest common divisor of all degrees of prime divisors.

There is a one-to-one correspondence between prime divisors P and places m_P (as defined earlier). In particular, the degree $\deg P$ of the prime divisor coincides with the degree of the place, i.e., the degree of the residue field extension $[\mathcal{O}_P/m_P : \mathbb{F}_q]$. We will denote the set of all prime divisors (closed points) of $\mathcal{X}$ by $\mathbb{P}_\mathcal{X}$.

For a divisor D of $\mathcal{X}$, we define a vector space $\mathcal{L}(D) := \{f \in k(\mathcal{X})^* : (f) + D \geq 0\} \cup \{0\}$, known as the *Riemann-Roch space*. If $\deg(D) < 0$, then $\mathcal{L}(D) = 0$, and $l(D) := \dim_k \mathcal{L}(D)$, is known as the *dimension* of the Riemann-Roch space. Moreover, the dimension $l(D)$ depends only on the class $[D]$, so we write $l([D]) := l(D)$.

Before we define a canonical divisor, we require the definition of a differential. For an affine open set $U \subseteq \mathcal{X}$, a *regular differential form* on U is a finite sum $\sum f_i\, dg_i$ where f_i, g_i are regular functions on U, and the symbols df satisfy the relations, $d(f + g) = df + dg$, $d(fg) = f\, dg + g\, df$, $da = 0$ for $a \in k$. The set of regular differentials on U is a $k[U]$-module, denoted $\Omega(U)$. Consider pairs (U, ω) where $U \subseteq \mathcal{X}$ is a non-empty affine open set and ω is a regular differential on U. Two pairs (U, ω) and (V, η) are equivalent if $\omega|_{U \cap V} = \eta|_{U \cap V}$. An equivalence class is called a *rational differential form* on $\mathcal{X}$. The $k(\mathcal{X})$-vector space of all rational differential forms is denoted $\Omega(\mathcal{X})$. It is one-dimensional over $k(\mathcal{X})$. For any local parameter t at a point P, every differential ω can be written near P as $\omega = f\, dt$ with $f \in k(\mathcal{X})$.

Let $\omega \in \Omega(\mathcal{X})$ be a non-zero differential. For a point $P \in \mathcal{X}$, choose a local parameter t_P and write $\omega = f_P\, dt_P$. The order of ω at P is $v_P(\omega) := v_P(f_P)$, where v_P is the valuation on $k(\mathcal{X})$ defined by $\mathcal{O}_P$. The *divisor of* ω is $(\omega) := \sum_{P \in \mathbb{P}_\mathcal{X}} v_P(\omega)\, P$. A divisor of the form (ω) is called a *canonical divisor*. Any two canonical divisors are linearly equivalent; their equivalence class is the *canonical class*, usually denoted by W.

Theorem 1.3 (Riemann-Roch, [8], Theorem 1.5.15)**.** *Let D be a divisor on the curve $\mathcal{X}$ of genus g. Then*

$$l(D) = \deg(D) + 1 - g + l(W - D).$$

The Riemann-Roch theorem is a fundamental result in Algebraic Geometry, and central in the construction of Algebraic Geometry codes. It will also be used to derive the functional equation of the zeta function of the curve $\mathcal{X}$ in the following section.

For the curve $\mathcal{X}$, the *genus* g is defined as $g = l(W)$ for any canonical divisor W. For a smooth projective curve of degree d in $\mathbb{P}^2$, this coincides with the Plücker formula $g = \frac{(d-1)(d-2)}{2}$. For a canonical divisor W of $\mathcal{X}$ of genus g, we have $\deg(W) = 2g - 2$, and for a divisor D of $\deg D > 2g - 2$, we have that $l(D) = \deg D - g + 1$ (since a canonical divisor W has degree $2g - 2$, so $l(W - C) = 0$).

Now we are ready to define the zeta function of an algebraic curve.

1.4.2 The Hasse-Weil zeta function

Definition 1.11. The Hasse-Weil zeta function of an algebraic curve is defined as the generating function

$$Z_\mathcal{X}(T) = \sum_{i \geq 0} A_i T^i, \tag{1.8}$$

where $A_i = |\{A \in \mathrm{Div}(\mathcal{X}) \mid A \geq 0 \text{ and } \deg(A) = i\}|$, i.e., the number of effective divisors of degree i.

The following lemma ensures that A_i is finite, so the power series (1.8) is well-defined.

Lemma 1.1. For every integer $n \geq 0$, there are only finitely many effective divisors of degree n.

Proof. An effective divisor is a finite sum of prime divisors. Therefore it suffices to show that the set $S := \{P \in \mathbb{P}_\mathcal{X} \mid \deg P \leq n\}$ is finite. Choose a non-constant rational function $x \in k(\mathcal{X})$. Consider the set $S_0 := \{Q \in \mathbb{P}_{\mathbb{F}_q(x)} \mid \deg Q \leq n\}$, where $\mathbb{P}_{\mathbb{F}_q(x)}$ denotes the set of places (prime divisors) of the rational function field $\mathbb{F}_q(x)$. For every prime divisor $P \in S$, its restriction $Q = P \cap \mathbb{F}_q(x)$ is a prime divisor of $\mathbb{F}_q(x)$ with $\deg Q \leq \deg P \leq n$, so $Q \in S_0$. Each $Q \in S_0$ has only finitely many extensions to $k(\mathcal{X})$ (since $[k(\mathcal{X}) : \mathbb{F}_q(x)] < \infty$). Thus, the finiteness of S follows from the finiteness of S_0. The set S_0 is finite because the places of $\mathbb{F}_q(x)$ of degree at most n are the point at infinity (degree 1), and the places corresponding to irreducible monic polynomials in $\mathbb{F}_q[x]$ of degree m for each $m \leq n$. There are only finitely many such irreducible polynomials, hence only finitely many places. □

Proposition 1.1 ([8], Proposition 5.1.6)**.** *The power series $Z_\mathcal{X}(T) = \sum_{i \geq 0}^{\infty} A_i T^i$ is convergent for $|T| < q^{-1}$. More precisely, we have for $|T| < q^{-1}$:*

1. *If $\mathcal{X}$ has genus $g = 0$ then*

$$Z_\mathcal{X}(T) = \frac{1}{q-1}\left(\frac{q}{1-(qT)^\partial} - \frac{1}{1-T^\partial}\right).$$

2. *If $g \geq 1$, then $Z_{\mathcal{X}}(T) = F(T) + G(T)$ with*

$$F(T) = \frac{1}{q-1} \sum_{0 \leq \deg[C] \leq 2g-2} q^{l([C])} \cdot T^{\deg[C]},$$

where the sum runs over all divisor classes $[C] \in \mathrm{Cl}(\mathcal{X})$ satisfying $0 \leq \deg[C] \leq 2g-2$, and

$$G(T) = \frac{h}{q-1} \left(q^{1-g} (qT)^{2g-2+\partial} \frac{1}{1-(qT)^{\partial}} - \frac{1}{1-T^{\partial}} \right).$$

To study the behavior of the zeta function of $\mathcal{X}$ under finite constant field extensions, it is convenient to have a second representation of $Z_{\mathcal{X}}(T)$ as an infinite product. We recall that an infinite product $\prod_{i=0}^{\infty}(1+a_i)$ (with complex numbers $a_i \neq -1$) is said to be *convergent* with limit $a \in \mathbb{C}$ if $\lim_{n\to\infty} \prod_{i=1}^{n}(1+a_i) = a \neq 0$. The product is called *absolutely convergent* if $\sum_{i=1}^{\infty} < \infty$. From analysis, we know that absolute convergence implies convergence of the product, and the limit of an absolutely convergent product is independent of the order of the factors. Moreover, if the product $\prod_{i=1}^{\infty}(1+a_i) = a$ is absolutely convergent, then $\prod_{i=1}^{\infty}(1+a_i)^{-1}$ converges absolutely too, and $\prod_{i=1}^{\infty}(1+a_i)^{-1} = a^{-1}$.

Proposition 1.2 (Euler Product)**.** *For $|T| < q^{-1}$, the zeta function can be represented as an absolutely convergent product*

$$Z_{\mathcal{X}}(T) = \prod_{P \in \mathbb{P}_{\mathcal{X}}} (1 - T^{\deg P})^{-1}. \tag{1.9}$$

In particular, $Z(T) \neq 0$ for $|T| < q^{-1}$.

Proof. The right-hand side of (1.9) converges absolutely for $|T| < q^{-1}$ since $\sum_{P \in \mathbb{P}_{\mathcal{X}}} |T|^{\deg P} \leq \sum_{D \geq 0} |T|^{\deg D} = \sum_{i=0}^{\infty} A_i |T|^i < \infty$ by Proposition 1.1. Each factor of (1.9) can be written as a geometric series, and we obtain

$$\prod_{P \in \mathbb{P}_{\mathcal{X}}} (1 - T^{\deg P})^{-1} = \prod_{P \in \mathbb{P}_{\mathcal{X}}} \sum_{n=0}^{\infty} T^{\deg(nP)}.$$

Expanding the product gives a sum over all families of non-negative integers n_P for $P \in \mathbb{P}_{\mathcal{X}}$ with only finitely many $n_P \neq 0$:

$$\prod_{P \in \mathbb{P}_{\mathcal{X}}} \sum_{n=0}^{\infty} T^{\deg(nP)} = \sum_{n_P} T^{\sum_P n_P \deg P}.$$

Every effective divisor D has a unique representation $D = \sum_P n_P P$ with such a family (n_P). Moreover, $\deg D = \sum_P n_P \deg P$. Hence the right-hand side is exactly $\sum_{D \geq 0} T^{\deg D}$. By definition $Z_{\mathcal{X}}(T) = \sum_{D \geq 0} T^{\deg D}$ (since grouping by degree yields $\sum_{n=0}^{\infty} A_n T^n$). Therefore

$$\prod_{P \in \mathbb{P}_{\mathcal{X}}} (1 - T^{\deg P})^{-1} = Z_{\mathcal{X}}(T),$$

and the equality holds for $|T| < q^{-1}$. □

Definition 1.12. The zeta function of an algebraic curve $\mathcal{X}$ is defined as

$$Z_{\mathcal{X}}(T) = \exp\left(\sum_{r=1}^{\infty} \frac{N_r}{r} T^r \right), \tag{1.10}$$

where N_r is defined as the number of $\mathbb{F}_{q^r}$-rational points of $\mathcal{X}$. This definition is equivalent to the other two definitions given by (1.8) and (1.9). To prove the equivalence of all three definitions, we require the following lemma.

Lemma 1.2 ([8], Lemma 5.1.9 (d))**.** Let $P \in \mathbb{P}_F$ be a place of degree m. Then $\mathrm{Con}_{F_r/F}(P) = P_1 + \ldots + P_d$ with $d := \gcd(m, r)$ pairwise distinct places $P_i \in \mathbb{P}_{F_r}$ and $\deg P_i = m/d$.

In Lemma 1.2, $F = k(\mathcal{X})$ is the function field of $\mathcal{X}$ and $F_r = k(\mathcal{X}_r)$ is the function field of the constant field extension $\mathcal{X}_r = \mathcal{X} \times_{\mathbb{F}_q} \mathbb{F}_{q^r}$. The set $\mathbb{P}_F$ corresponds to the set $\mathbb{P}_\mathcal{X}$ of prime divisors (closed points) of $\mathcal{X}$. A place $P \in \mathbb{P}_F$ of degree m corresponds to a prime divisor $P \in \mathbb{P}_\mathcal{X}$ of degree m. The conorm $\mathrm{Con}_{F_r/F}(P)$ in function field language translates to the pullback of the divisor P under the finite morphism $\mathcal{X}_r \to \mathcal{X}$. In geometric terms, it is the sum of the distinct closed points of $\mathcal{X}_r$ lying over P, each taken with multiplicity 1. Thus, the lemma says that if P has degree m, its pullback splits into $d = \gcd(m, r)$ distinct points $P_1, \ldots, P_d$ on $\mathcal{X}_r$, each of degree m/d.

Proposition 1.3. *The two definitions of the zeta function of an algebraic curve, given by* (1.9) *and* (1.10) *are equivalent.*

Proof. Upon taking logarithm on both sides of (1.9), we get

$$\log Z_\mathcal{X}(T) = \log\left(\prod_{P \in \mathbb{P}_\mathcal{X}} (1 - T^{\deg P})^{-1}\right).$$

We can expand the right-hand side as a power series to get

$$\log\left(\prod_{P \in \mathbb{P}_\mathcal{X}} (1 - T^{\deg P})^{-1}\right) = \sum_{P \in \mathbb{P}_\mathcal{X}} \sum_{i=1}^{\infty} \frac{T^{i \deg P}}{i}. \tag{1.11}$$

Let $B_d = \#\{P \mid \deg P = d\}$. Then

$$\sum_{P \in \mathbb{P}_\mathcal{X}} T^{i \deg P} = \sum_{d=1}^{\infty} B_d T^{id}.$$

Upon substituting this in (1.11), we get that

$$\sum_{i=1}^{\infty} \frac{1}{i} \sum_{d=1}^{\infty} B_d T^{id} = \sum_{i,d \geq 1} \frac{B_d}{i} T^{id}.$$

Let $r = id$. For a fixed r, the pairs (i, d) with $id = r$ are those where $d \mid r$ and $i = r/d$. Therefore,

$$\log Z_\mathcal{X}(T) = \sum_{r=1}^{\infty} \left(\sum_{d \mid r} B_d \cdot \frac{1}{r/d}\right) T^r = \sum_{r=1}^{\infty} \frac{1}{r} \left(\sum_{d \mid r} d \cdot B_d\right) T^r.$$

From Lemma 1.2, the number of $\mathbb{F}_{q^r}$ rational points of $\mathcal{X}$ is $\sum_{d \mid r} d \cdot B_d = N_r$. Thus, we get

$$\log Z_\mathcal{X}(T) = \sum_{r=1}^{\infty} \frac{N_r}{r} T^r.$$

Exponentiation of both sides yields

$$Z_{\mathcal{X}}(T) = \exp\left(\sum_{r=1}^{\infty} \frac{N_r}{r} T^r\right).$$

□

Corollary 1.1 (F. K. Schmidt, [8], Corollary 5.1.7)**.** *For an algebraic curve $\mathcal{X}$ over $\mathbb{F}_q$, the greatest common divisor of the degrees of all divisors is* 1*, i.e., $\partial = 1$. Equivalently, there exists a divisor of degree* 1*.*

Corollary 1.2 ([8], Corollary 5.1.12)**.** *1. Every function field of $\mathcal{X}$ of genus* 0 *is rational, and its zeta function is*

$$Z_{\mathcal{X}}(t) = \frac{1}{(1-t)(1-qt)}.$$

2. If $\mathcal{X}$ has genus $g \geq 1$, its zeta function can be written in the form

$$Z_{\mathcal{X}}(t) = F(t) + G(t)$$

with

$$F(t) = \frac{1}{q-1} \sum_{0 \leq \deg[C] \leq 2g-2} q^{l([C])} \cdot t^{\deg[C]}$$

and

$$G(t) = \frac{h}{q-1}\left(q^g t^{2g-1} \frac{1}{1-qt} - \frac{1}{1-t}\right).$$

Proposition 1.4 (Functional equation of the zeta function)**.** *The zeta function of $\mathcal{X}$ satisfies the functional equation*

$$Z_{\mathcal{X}}(T) = q^{g-1} T^{2g-2} Z_{\mathcal{X}}\left(\frac{1}{qT}\right) \tag{1.12}$$

Proof. For $g = 0$, we use the formula of the zeta function from Corollary 1.2(1) to obtain

$$Z_{\mathcal{X}}\left(\frac{1}{qT}\right) = \frac{qT^2}{(1-qT)(1-T)}.$$

Upon multiplying both sides by $q^{-1}t^{-2}$, we obtain (1.12). For $g \geq 1$, we write $Z_{\mathcal{X}}(T) = F(T) + G(T)$ as in Corollary 1.2 (2). Let W be a canonical divisor of $\mathcal{X}$, then

$$\begin{aligned}
(q-1)F(T) &= \sum_{0 \leq \deg[C] \leq 2g-2} q^{l([C])} \cdot T^{\deg[C]} \\
&= \sum_{0 \leq \deg[C] \leq 2g-2} q^{\deg[C]+1-g+l([W-C])} \cdot T^{\deg[C]} \\
&= q^{g-1}T^{2g-2} \sum_{0 \leq \deg[C] \leq 2g-2} q^{\deg[C]-(2g-2)+l([W-C])} \cdot T^{\deg[C]-(2g-2)} \\
&= q^{g-1}T^{2g-2} \sum_{0 \leq \deg[C] \leq 2g-2} q^{l([W-C])} \cdot \left(\frac{1}{qT}\right)^{\deg[W-C]}
\end{aligned}$$

and we get

$$(q-1)F\left(\frac{1}{qT}\right) = q^{1-g}T^{2-2g}(q-1)F(T) \tag{1.13}$$

We used the Riemann-Roch theorem, the fact that $\deg[W] = 2g-2$ and, if $[C]$ runs over all divisor classes with $0 \leq \deg[C] \leq 2g-2$, so does $[W-C]$. For the function $G(T)$, we obtain

$$q^{g-1}T^{2g-2}G\left(\frac{1}{qT}\right) = \frac{h}{q-1}q^{g-1}T^{2g-2}\left(q^g\left(\frac{1}{qT}\right)^{2g-1}\frac{1}{1-q\frac{1}{qT}} - \frac{1}{1-\frac{1}{qT}}\right)$$

and thus,

$$q^{g-1}t^{2g-2}G\left(\frac{1}{qT}\right) = \frac{h}{q-1}\left(\frac{1}{T}\frac{1}{1-\frac{1}{T}} - \frac{q^gT^{2g-1}}{qT(1-\frac{1}{qT})}\right) = G(T). \tag{1.14}$$

Adding (1.13) and (1.14) yields the functional equation for $Z_{\mathcal{X}}(T)$. □

Definition 1.13 ([8], Definition 5.1.14)**.** The polynomial

$$L(T) := (1-T)(1-qT)Z_{\mathcal{X}}(T) \tag{1.15}$$

is called the *L-polynomial* of $\mathcal{X}$.

Remark 1.3. We shall refer to (1.15) as the *zeta polynomial* of $\mathcal{X}$, and denote it by $P_{\mathcal{X}}(T)$. By Corollary 1.2, it is obvious that $P_{\mathcal{X}}(T)$ is a polynomial of degree $\leq 2g$. We can also observe that $P_{\mathcal{X}}(T)$ contains all information about the numbers A_i in (1.8).

For A_i, the number of effective divisors of degree i on $\mathcal{X}$, we have that $A_i = 0$ if $i < 0$ because an effective divisor has non-negative degree. For $i > 2g-2$, Lemma 5.1.4(b) [8] tells us that in a fixed divisor class $[C]$ of degree i the number of effective divisors is $\frac{q^{l(C)}-1}{q-1}$. By the Riemann-Roch theorem 1.3, we have $l(C) = i+1-g$ because a canonical divisor has degree $2g-2$ and $l(K-C) = 0$. Moreover, from Proposition 5.1.3 [8] we have that there are exactly h divisor classes of degree i (the class number $h = \#\mathrm{Cl}^0(\mathcal{X})$). Consequently, the total number A_i of effective divisors of degree $i > 2g-2$ is $A_i = h\cdot\frac{q^{i+1-g}-1}{q-1}$. Thus, for $i \notin \{0,1,\ldots,2g\}$,

$$A_i - (q+1)A_{i-1} + qA_{i-2} = 0. \tag{1.16}$$

Let

$$p_i = A_i - (q+1)A_{i-1} + qA_{i-2}. \tag{1.17}$$

Upon multiplying both sides of (1.8) with $(1-T)(1-qT)$, we obtain

$$(1-T)(1-qT)Z_{\mathcal{X}}(T) = \sum_{i=0}^{\infty}A_iT^i - (q+1)\sum_{i=0}^{\infty}A_iT^{i+1} + q\sum_{i=0}^{\infty}A_iT^{i+2}$$

Re-indexing the sums, and collecting coefficients of T^i for each $i \geq 0$ gives us,

$$\begin{aligned}(1-T)(1-qT)Z_{\mathcal{X}}(T) &= \sum_{i=0}^{\infty}A_iT^i - (q+1)\sum_{i=1}^{\infty}A_{i-1}T^i + q\sum_{i=2}^{\infty}A_{i-2}T^i \\ &= \sum_{i=0}^{\infty}(A_i - (q+1)A_{i-1} + qA_{i-2})T^i,\end{aligned}$$

where $A_{-1} = A_{-2} = 0$ and the coefficient of T^i is precisely p_i from (1.17). From the Riemann-Roch theorem, we know that $p_i = 0$ for all $i > 2g$. Thus, the right-hand side is a polynomial of degree at most $2g$ denoted by $P_{\mathcal{X}}(T) = p_0 + p_1T + \cdots + p_{2g}T^{2g}$. Using this, the zeta function can be written as the rational

function

$$Z_{\mathcal{X}}(T) = \frac{P_{\mathcal{X}}(T)}{(1-T)(1-qT)}. \tag{1.18}$$

Pellikaan [18] defines a two-variable zeta function as the power series

$$Z_{\mathcal{X}}(T,u) = \sum_{[D]} \frac{u^{l(D)}-1}{u-1} T^{\deg D}, \tag{1.19}$$

and gives a decomposition

$$(u-1)Z_{\mathcal{X}}(T,u) = \sum_{d=0}^{2g-2} \sum_{\deg[D]=d} u^{l(D)} T^{\deg D} + \sum_{d>2g-2} h u^{d+1-g} T^d - \sum_{d\geq 0} hT^d. \tag{1.20}$$

Upon multiplying both sides of (1.19) by $(u-1)$, we get

$$(u-1)Z_{\mathcal{X}}(T,u) = \sum_{[D]} u^{l(D)} T^{\deg D} - \sum_{[D]} T^{\deg(D)}.$$

Since the degree of a divisor class is well-defined, we can group the sum by the degree $d = \deg D$. And there exists a bijection between $\mathrm{Cl}^0(\mathcal{X}) \to \{\text{divisor classes of degree d}\}$, $[C] \mapsto [C] + d[D_0]$, where $[D_0]$ is a divisor of degree 1 (it exists due to Corollary 1.1). Thus, the cardinality of the set of divisors of degree d equals $|\mathrm{Cl}^0(\mathcal{X})| = h$. Thus, we get

$$\sum_{[D]} T^{\deg D} = \sum_{d=0}^{\infty} hT^d, \tag{1.21}$$

and we get

$$\sum_{[D]} u^{l(D)} T^{\deg D} = \sum_{d=0}^{2g-2} \sum_{\deg D=d} u^{l(D)} T^d + \sum_{d>2g-2} \sum_{\deg D=d} u^{l(D)} T^d. \tag{1.22}$$

For $\deg D > 2g-2$, we know that $l(D) = d+1-g$. Thus, we get

$$\sum_{\deg D=d} u^{l(D)} = h u^{d+1-g} \text{ for d} \geq \text{2g-1}. \tag{1.23}$$

Upon combining (1.21), (1.22), and (1.23), we get (1.20). For a base field $\mathbb{F}_q$ of q elements and for $u = q$, the two-variable zeta function agrees with the Hasse-Weil zeta function: $Z_{\mathcal{X}}(T,q) = Z_{\mathcal{X}}(T)$.

Theorem 1.4 ([8], Theorem 5.1.15). *1. $P_{\mathcal{X}}(T) \in \mathbb{Z}[T]$ and $\deg P_{\mathcal{X}}(T) = 2g$.*

2. (Functional equation) $P_{\mathcal{X}}(T) = q^g T^{2g} P_{\mathcal{X}}\left(\frac{1}{qT}\right)$.

3. $P_{\mathcal{X}}(1) = h$, the class number of $\mathcal{X}$.

4. Write $P_{\mathcal{X}}(T) = a_0 + a_1 T + \cdots + a_{2g} T^{2g}$. Then the following hold:

(a) $a_0 = 1$ and $a_{2g} = q^g$.

(b) $a_{2g-i} = q^{g-i} a_i$ for $0 \leq i \leq g$.

(c) $a_1 = N - (q+1)$, where N is the number of $\mathbb{F}_q$-rational points of $\mathcal{X}$.

5. $P_{\mathcal{X}}(T)$ factors in $\mathbb{C}[T]$ as

$$P_{\mathcal{X}}(T) = \prod_{i=1}^{2g} (1 - \alpha_i T),$$

where the α_i are algebraic integers. They can be arranged so that $\alpha_i\alpha_{g+i} = q$ for $i = 1, \ldots, g$.

6. *If $P_r(T) := (1-T)(1-q^rT)Z_{\mathcal{X}_r}(T)$ denotes the zeta polynomial of the constant field extension $\mathcal{X}_r = \mathcal{X} \times_{\mathbb{F}_q} \mathbb{F}_{q^r}$, then*

$$P_r(T) = \prod_{i=1}^{2g}(1-\alpha_i^rT),$$

where the α_i are as in (e).

Proof. For $g \geq 0$, all the assertions are trivial, hence we assume from now on that $g \geq 1$. (1) follows from (1.18) and (4), and (2) follows from Proposition 1.12. (3) follows by substituting $T = 1$ in Corollary 1.2 (2). To prove (4), write $P_{\mathcal{X}}(T) = a_0 + a_1T + \cdots + a_{2g}T^{2g}$. The functional equation (2) gives

$$P_{\mathcal{X}}(T) = q^gT^{2g}P_{\mathcal{X}}\left(\frac{1}{qT}\right) = q^gT^{2g}\left(a_0 + a_1\frac{1}{qT} + a_2\frac{1}{q^2T^2} + \cdots + a_{2g}\frac{1}{q^{2g}T^{2g}}\right).$$

Multiplying out, the coefficient of T^{2g-i} on the right-hand side is $\frac{a_i}{q^i} \cdot q^g = q^{g-i}a_i$. Hence

$$P_{\mathcal{X}}(T) = \frac{a_{2g}}{q^g} + \frac{a_{2g-1}}{q^{g-1}}T + \cdots + \frac{a_1}{q^1}T^{2g-1} + q^ga_0T^{2g}.$$

Comparing coefficients with $P_{\mathcal{X}}(T) = \sum_{i=0}^{2g} a_iT^i$ yields

$$a_{2g-i} = q^{g-i}a_i \qquad (0 \leq i \leq g). \tag{2}$$

Now recall that $P_{\mathcal{X}}(T) = (1-T)(1-qT)Z_{\mathcal{X}}(T)$, where $Z_{\mathcal{X}}(T) = \sum_{i=0}^{\infty} A_iT^i$ and A_i is the number of effective divisors of degree i. Expanding the product,

$$P_{\mathcal{X}}(T) = \left(1-(q+1)T+qT^2\right)\sum_{i=0}^{\infty}A_iT^i = \sum_{i=0}^{\infty}A_iT^i - (q+1)\sum_{i=0}^{\infty}A_iT^{i+1} + q\sum_{i=0}^{\infty}A_iT^{i+2}.$$

The constant term is $A_0 = 1$, so $a_0 = 1$. The coefficient of T^1 is $A_1 - (q+1)A_0 = N - (q+1)$, because $A_1 = N$. Hence $a_1 = N - (q+1)$. Finally, from (2) with $i = 0$ we get $a_{2g} = q^ga_0 = q^g$.For a detailed proof of the rest, one can refer to [8]. □

The above theorem shows that the number of $\mathbb{F}_q$-rational points N can be easily calculated if the zeta polynomial of $\mathcal{X}$ is known. More generally, we consider the number N_r, the number of $\mathbb{F}_{q^r}$-rational points of $\mathcal{X}$, i.e., the number of points of $\mathcal{X}$ with coordinates in $\mathbb{F}_{q^r}$. The following result explicitly gives the relation between the reciprocal zeros of the zeta polynomial of $\mathcal{X}$ and its $\mathbb{F}_q$-rational points.

Corollary 1.3. *For every integer $r \geq 1$, the number N_r satisfies*

$$N_r = q^r + 1 - \sum_{i=1}^{2g}\alpha_i^r, \tag{1.24}$$

where $\alpha_1, \ldots, \alpha_{2g} \in \mathbb{C}$ are the reciprocals of the zeros of the zeta polynomial $P_{\mathcal{X}}(T)$, i.e., $P_{\mathcal{X}}(T) = \prod_{i=1}^{2g}(1-\alpha_iT)$. In particular, for $r = 1$ we have

$$N = q + 1 - \sum_{i=1}^{2g}\alpha_i. \tag{1.25}$$

Proof. By Theorem 1.4(4), the coefficient of T^1 in the zeta polynomial $P_{\mathcal{X}}(T)$ equals $N_1 - (q+1)$. And for the constant field extension $\mathcal{X}_r = \mathcal{X} \times_{\mathbb{F}_q} \mathbb{F}_{q^r}$, the same theorem gives that the coefficient of T^1 in its zeta polynomial $P_{\mathcal{X}_r}(T)$ equals $N_r - (q^r + 1)$. On the other hand, part (6) of Theorem 1.4 gives us that

$$P_{\mathcal{X}_r}(T) = \prod_{i=1}^{2g} (1 - \alpha_i^r T).$$

Upon expanding the product, the coefficient of T^1 is $-\sum_{i=1}^{2g} \alpha_i^r$. Equating the two expressions yields $N_r - (q^r + 1) = -\sum_{i=1}^{2g} \alpha_i^r$, which gives us (1.24). □

1.4.3 The Riemann hypothesis analogue for curves

Corollary 1.3 tells us that the numbers N_r of rational points in extensions are given by $q^r + 1 - \sum_{i=1}^{2g} \alpha_i^r$. So the α_i control how far N_r is from $q^r + 1$. If we can show that $|\alpha_i| = \sqrt{q}$ for every i, then we would immediately obtain the Hasse–Weil bound $|N_r - (q^r + 1)| \leq 2gq^{r/2}$. This bound will be discussed in more detail later. The main result of this section is

Theorem 1.5 (Hasse-Weil, [8], Theorem 5.2.1)**.** *The reciprocals of the roots of $P_{\mathcal{X}}(T)$ satisfy $|\alpha_i| = \sqrt{q}$ for $i = 1, \ldots, 2g$.*

The *Hasse-Weil* theorem is often referred to as the *Riemann hypothesis for algebraic curves over finite fields*. We can understand this notation by looking at the zeta function $Z_{\mathcal{X}}(T)$ of the curve as an analogue of the classical Riemann-zeta function

$$\zeta(s) := \sum_{n=1}^{\infty} n^{-s}$$

where $s \in \mathbb{C}$ and $\mathrm{Re}(s) > 1$ in the following way. Define the absolute norm of a divisor $A \in \mathrm{Div}(\mathcal{X})$ by $\mathcal{N}(A) := q^{\deg A}$. For example, the absolute norm $\mathcal{N}(P)$ of a prime divisor $P \in \mathbb{P}_{\mathcal{X}}$ is the cardinality of its residue field k_P. Then the function

$$\zeta_{\mathcal{X}}(s) := Z_{\mathcal{X}}(q^{-s}) = \sum_{n=0}^{\infty} A_n q^{-sn} = \sum_{A \geq 0} \mathcal{N}(A)^{-s}$$

is the analogue of the classical Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$. It is well known that $\zeta(s)$ extends to a meromorphic function on $\mathbb{C}$. The Riemann Hypothesis states that all non-trivial zeros of $\zeta(s)$ lie on the line $\mathrm{Re}(s) = 1/2$.

In the case of curves over finite fields, the Hasse-Weil theorem 1.5 implies that

$$\zeta_{\mathcal{X}}(s) = 0 \implies Z_{\mathcal{X}}(q^{-s}) = 0 \implies |q^{-s}| = q^{-1/2}.$$

Since $|q^{-s}| = q^{-\mathrm{Re}(s)}$, we obtain $\mathrm{Re}(s) = 1/2$ for every zero of $\zeta_{\mathcal{X}}(s)$. Thus, the Hasse-Weil theorem is the exact analogue of the classical Riemann Hypothesis for algebraic curves over finite fields.

Using the Hasse-Weil theorem 1.5, we can obtain an important conclusion.

Theorem 1.6 (Hasse-Weil bound, [8], Theorem 5.2.3)**.** *The number of $\mathbb{F}_q$ rational points of $\mathcal{X}$ satisfies the inequality*

$$|N - (q+1)| \leq 2g\sqrt{q}.$$

Proof. From Corollary 1.3, we have $N - (q+1) = -\sum_{i=1}^{2g} \alpha_i$. And from the Hasse-Weil theorem 1.5 we have that $|\alpha_i| = \sqrt{q}$ for every i. Hence $|N - (q+1)| \leq \sum_{i=1}^{2g} |\alpha_i| = 2g\sqrt{q}$. □

The zeta function of an algebraic curve can be expanded as

$$\frac{P_\chi(T)}{(1-T)(1-qT)} = \prod_{m=1}^{\infty}(1-T^m)^{-B_m}, \tag{1.26}$$

where the B_m's are non-negative integers (since they are the number of places (closed points) of degree exactly m on the curve. Thus, the zeta functions of algebraic curves satisfy a positive Weil system.

Definition 1.14 (Tsfasman-Vladuts [15])**.** For a given q, a subset $\Omega = \{\omega_1, \ldots, \omega_{2g}\}$ of complex numbers is called a Weil system if

1. ω_i is an algebraic integer,
2. all conjugates of ω_i belong to Ω,
3. if ω_i is real, its multiplicity in Ω is even,
4. $|\omega_i| = \sqrt{q}$.
   Let
   $$\frac{(1-\omega_1 T)\cdots(1-\omega_{2g}T)}{(1-T)(1-qT)} = \prod_{m=1}^{\infty}(1-T^m)^{-B_m}.$$
   A Weil system is called positive if
5. $B_m \geq 0$

The fact that condition (4) holds for any curve is known as the Riemann hypothesis analogue for algebraic curves over a finite field. Condition (5) is trivial for algebraic curves. Using the "explicit formulas" method (from analytic number theory) together with the positive Weil system property, we get the following upper bound on the number of rational points N relative to the genus g.

Theorem 1.7 (Drinfeld-Vladtus [19])**.** *For an infinite family of algebraic curves of increasing genus g over a field of q elements,*

$$\limsup_{g\to\infty} N/g \leq \sqrt{q} - 1. \tag{1.27}$$

Chapter 2

Zeta functions of Hamming metric codes

In this chapter, we will give three definitions for the zeta functions of Hamming metric codes, all due to Duursma. We will then calculate its degree, prove its existence and uniqueness, obtain its functional equation, and define some of its properties. Following that, we will discuss the Riemann hypothesis analogue for Hamming metric codes, with a focus on self-dual linear codes and Duursma's conjecture about extremal self-dual linear codes.

2.1 First definition

In 1999 [5], Iwan Duursma defined the zeta function for a linear code in the Hamming metric, as a generating function of its Hamming weight enumerator. The following definition captures this idea.

Definition 2.1. A polynomial $P_C(T)$ for which

$$\frac{(xT + (1-T)y)^n}{(1-T)(1-qT)} P_C(T) = \cdots + \frac{A_C(x,y) - x^n}{q-1} T^{n-d} + \cdots \tag{2.1}$$

is called a *Duursma zeta polynomial* of the linear code C. The quotient

$$Z_C(T) = \frac{P_C(T)}{(1-T)(1-qT)}$$

is called the *zeta function of the linear code*. It has a structure analogous to the zeta function (1.18) of an algebraic curve.

In order to understand the expansion of (2.1) and to obtain the second definition of the zeta polynomial, we require the following definition.

Definition 2.2. Define c_j by

$$\frac{(xT + (1-T)y)^n}{(1-T)(1-qT)} = \sum_{r=0}^{\infty} c_r(x,y) T^r. \tag{2.2}$$

Define $M_{n,\delta}$ by

$$M_{n,\delta}(x,y) = x^n + (q-1)c_{n-\delta}(x,y).$$

This is called the *MDS virtual weight enumerator of length n and distance δ*. We can see that

$$\frac{1}{(1-T)(1-qT)} = \left(\sum_{n=0}^{\infty} T^n\right)\left(\sum_{m=0}^{\infty} q^m T^m\right) = \sum_{r=0}^{\infty} \frac{q^{r+1}-1}{q-1} T^r$$

and

$$(xT + (1-T)y)^n = \sum_{i=0}^{n} \binom{n}{i} y^{n-i}(x-y)^i T^i = \sum_{i=0}^{n} \sum_{j=0}^{i} \binom{n}{i}\binom{i}{j} y^{n-j} x^j T^i (-1)^{i-j}.$$

Therefore,

$$c_r(x,y) = \sum_{i+j=r} \frac{q^{j+1}-1}{q-1} \binom{n}{i} y^{n-i}(x-y)^i. \tag{2.3}$$

Using the expansion of coefficients $c_r(x,y)$ in (2.3), we can understand the expansion of (2.1).

Lemma 2.1. If we expand $\frac{(xT+y(1-T))^n}{(1-T)(1-qT)}$ in powers of T, we find that it is equal to

$$b_{0,0}y^n T^0 + \left(b_{1,1}xy^{n-1} + b_{1,0}y^n\right)T^1 + \left(b_{2,2}x^2y^{n-2} + b_{2,1}xy^{n-1} + b_{2,0}y^n\right)T^2 + \cdots +$$
$$\left(b_{n-d,n-d}x^{n-d}y^d + b_{n-d,n-d-1}x^{n-d-1}y^{d+1} + \cdots + b_{n-d,0}y^n\right)T^{n-d} + \cdots$$

where $b_{i,j}$ are the coefficients given by

$$b_{r,l} = \sum_{i=l}^{r} \frac{q^{r-i+1}-1}{q-1}\binom{n}{i}\binom{i}{l}(-1)^{i-l}$$

for $0 \le l \le r \le n-d$ and $b_{r,l} = 0$ otherwise.

Proof. Upon expanding $(x-y)^i$ in (2.3) using the binomial theorem, we get,

$$(x-y)^i = \sum_{l=0}^{i} \binom{i}{l} x^l (-y)^{i-l} = \sum_{l=0}^{i} \binom{i}{l}(-1)^{i-l}x^l y^{i-l}.$$

Substituting this back into (2.3) gives us

$$c_r(x,y) = \sum_{i+j=r} \sum_{l=0}^{i} \frac{q^{j+1}-1}{q-1}\binom{n}{i}\binom{i}{l}(-1)^{i-l}x^l y^{n-l}.$$

Now collect by powers of x (i.e., by l). For a fixed r and l, the coefficient of $x^l y^{n-l}$ in $c_r(x,y)$ is:

$$\sum_{i=l}^{r}\binom{i}{l}(-1)^{i-l}\binom{n}{i}\frac{q^{r-i+1}-1}{q-1}.$$

Since $j = r-i$, we have $q^{j+1} = q^{r-i+1}$. Define the coefficients $b_{r,l}$ by collecting terms

$$b_{r,l} = \sum_{i=l}^{l} \frac{q^{r-i+1}-1}{q-1}\binom{n}{i}\binom{i}{l}(-1)^{i-l}$$

for $0 \le l \le r \le n-d$ and $b_{r,l} = 0$ otherwise, since there are no valid indices i satisfying both $l \le i$ and $i \le r$. □

Proposition 2.1 ([2], Proposition 1)**.** *Let the left-hand side of* (2.1) *with* $P_C(T) = 1$ *be the generating function for a sequence of weight enumerators* $M_{n,d}$ *of length* n *and minimum distance* d*,*

$$\frac{1}{(1-T)(1-qT)}(y(1-T)+xT)^n = \cdots + \frac{M_{n,d}(x,y) - x^n}{q-1}T^{n-d} + \cdots.$$

The weight enumerators $M_{n,d}$ are MDS. Equivalently, the weight enumerators $M_{n,d}$ and $M_{n,n+2-d}$ are in duality under the Mac Williams transform (1.2).

Following Duursma [5], the degree of the zeta polynomial $P_C(T)$ is given by $\deg P_C = \gamma + \gamma^\perp$, where $\gamma = n + 1 - k - d$ and $\gamma^\perp = n + 1 - k^\perp - d^\perp$ are the genus of C and its dual. Substituting $k^\perp = n - k$ yields $\deg P_C = n + 2 - d - d^\perp$. The duality relation of the zeta polynomial is given by

$$P_{C^\perp}(T) = P_C(\frac{1}{qT})q^\gamma T^{\gamma+\gamma^\perp}$$

where γ is the genus of the code C of length n and minimum distance d and $\gamma^\perp$ is the genus of its dual. Duality for zeta functions is given by

$$Z_{C^\perp}(T) = Z_C(1/qT)q^{\gamma-1}T^{\gamma+\gamma^\perp-2}. \tag{2.4}$$

In addition to these properties, for a self-dual linear code, we also have that $P_{C^\perp}(T) = P_C(T)$, and hence $Z_C(T) = Z_{C^\perp}(T)$. The following proposition justifies the existence of the zeta polynomial given by (2.1).

Proposition 2.2. *The Duursma zeta polynomial P_C exists and is unique, provided that $d^\perp \geq 2$.*

Proof. Let C be an $[n, k, d]_q$ code with weight enumerator polynomial $A_C = \sum_{i=0}^{n} A_i x^{n-i} y^i$. Using Lemma 2.1, we can expand

$$L(T) = \frac{(xT + (1-T)y)^n}{(1-T)(1-qT)} = \sum_{m=0}^{\infty}\left(\sum_{l=0}^{m} b_{m,l}x^l y^{n-l}\right)T^m$$

such that

$$b_{m,l} = \sum_{i=l}^{m} \frac{q^{m-i+1}-1}{q-1}\binom{n}{i}\binom{i}{l}(-1)^{i-l}$$

for $0 \leq l \leq m \leq n-d$, $b_{m,l} = 0$ otherwise, and $b_{m,m} = \frac{q^{m-m+1}-1}{q-1}\binom{n}{m}\binom{m}{m} = \binom{n}{m} \neq 0$.

From [5], we know that $\deg P_C = n + 2 - d - d^\perp$. If we assume that $d^\perp \geq 2$, then we get that $\deg P_C \leq n - d$. Thus, we can write $P_C(T) = a_0 + a_1 T + \cdots + a_{n-d}T^{n-d}$. Next, we get the coefficient of T^{n-d} in the product $L(T) \cdot P_C(T)$ to be

$$\sum_{j=0}^{n-d} a_j \left(\sum_{l=0}^{n-d-j} b_{n-d-j,l}x^l y^{n-l}\right) = \sum_{l=0}^{n-d}\left(\sum_{j=0}^{n-d-l} a_j b_{n-d-j,l}\right)x^l y^{n-l}. \tag{2.5}$$

We can change the index by substituting $s = n - d - j$ and simplifying to get

$$\sum_{j=0}^{n-d-l} a_j b_{n-d-j,l} = \sum_{s=l}^{n-d} a_{n-d-s}b_{s,l}. \tag{2.6}$$

Upon substituting (2.6) in (2.5), we get

$$\sum_{l=0}^{n-d}\left(\sum_{s=l}^{n-d} a_{n-d-s}b_{s,l}\right)x^l y^{n-l}. \tag{2.7}$$

Next, on the right-hand side of Definition 2.1, we have

$$\frac{A_C(x,y)-x^n}{q-1}=\sum_{i=1}^{n}\frac{A_i x^{n-i}y^i}{q-1}.$$

To rewrite this in a form similar to (2.7), we can substitute $l=n-i$ to get

$$\sum_{l=0}^{n-1}\frac{A_{n-l}}{q-1}x^l y^{n-l}.$$

Now, we match the coefficients of T^{n-d} on the left-hand side and right-hand side. On the left-hand side, in the coefficient of T^{n-d}, l goes up to $n-d$. For $l>n-d$, the coefficient is 0. And on the right-hand side, the coefficient is $\frac{A_{n-l}}{q-1}$, but for $l>n-d$, we get $n-l<d$, and in this case $A_{n-l}=0$. So the coefficient on the right-hand side is also zero. Thus, we use the equations from $l=0$ to $l=n-d$ to form the matrix equation $B\cdot\vec{a}=\vec{A}$ of coefficients of $x^l y^{n-l}$ given by

$$\begin{pmatrix} b_{0,0} & b_{1,0} & \dots & b_{n-d,0} \\ 0 & b_{1,1} & \dots & b_{n-d,1} \\ 0 & 0 & b_{2,2} & \dots \\ \vdots & & \ddots & \vdots \\ 0 & \dots & 0 & b_{n-d,n-d} \end{pmatrix}\begin{pmatrix} a_{n-d} \\ a_{n-d-1} \\ \vdots \\ a_0 \end{pmatrix}=\begin{pmatrix} A_n/(q-1) \\ A_{n-1}/(q-1) \\ \vdots \\ A_d/(q-1) \end{pmatrix}.$$

From Lemma 2.1 above, we know that the diagonal entries of this matrix are binomial coefficients, $b_{i,j}=\binom{n}{i}$, hence are nonzero. Therefore, the matrix is invertible, and the existence and uniqueness are established. □

Remark 2.1.

1. It is clear from the proof of Proposition 2.2 that to calculate the zeta polynomial of a code, we require the coefficients of its weight enumerator function, and the coefficients of $L(T)$, which can be computed using Lemma 2.1.

2. In (2.1), let $P_C(T)=a_0+a_1T+\cdots+a_{n-d}T^{n-d}$ be the zeta polynomial of degree at most $n-d$, and let $L(T)=\frac{(xT+(1-T)y)^n}{(1-T)(1-qT)}=c_0+c_1T+c_2T^2+\cdots$. Then upon multiplying $P_C(T)\cdot L(T)$, we get the coefficient of T^r to be $\sum_{\substack{i+j=r\\0\le i\le n-d\\ j\ge 0}} a_i c_j(x,y)$. And the coefficient of T^{n-d} will be $\sum_{i=0}^{n-d} a_i c_{n-d-i}(x,y)$.
   We need to choose a zeta polynomial $P_C(T)$ in such a way that

$$\sum_{i=0}^{n-d}a_i c_{n-d-i}(x,y)=\frac{A_C(x,y)-x^n}{q-1}=\sum_{i=0}^{n}\frac{A_i x^{n-i}y^i-x^n}{q-1}=\sum_{i=1}^{n}\frac{A_i x^{n-i}y^i}{q-1}.$$

3. Using Proposition 2.2, we can refine Definition 2.1, and define the zeta polynomial $P_C(T)$ as a polynomial of degree at most $n-d$ such that the generating function

$$\frac{P_C(T)}{(1-T)(1-qT)}(y(1-T)+xT)^n$$

has expansion

$$\cdots+\frac{A(x,y)-x^n}{q-1}T^{n-d}+\cdots.$$

The following proposition gives us a way to calculate the weight distribution of a linear code, without

computing the whole weight enumerator of the code.

Proposition 2.3. *If $d^\perp \geq 2$, then we get the following generalized equation to calculate the weight distribution of a code C*

$$A_{d+i} = \binom{n}{n-d-i} \sum_{j=0}^{i} a_j \left[\sum_{t=0}^{i-j} \left(q^{i-j-t+1} - 1 \right) \binom{d+i}{t} (-1)^t \right]. \tag{2.8}$$

Proof. Recall the matrix equation from the proof of Proposition 2.2 given by

$$\begin{pmatrix} b_{0,0} & b_{1,0} & \dots & b_{n-d,0} \\ 0 & b_{1,1} & \dots & b_{n-d,1} \\ 0 & 0 & b_{2,2} & \dots \\ \vdots & & \ddots & \vdots \\ 0 & \dots & 0 & b_{n-d,n-d} \end{pmatrix} \begin{pmatrix} a_{n-d} \\ a_{n-d-1} \\ \vdots \\ a_0 \end{pmatrix} = \begin{pmatrix} A_n/(q-1) \\ A_{n-1}/(q-1) \\ \vdots \\ A_d/(q-1) \end{pmatrix}.$$

For each row l, ranging from 0 to $n-d$, we have

$$\sum_{j=0}^{n-d-l} a_j b_{n-d-j,l} = \frac{A_{n-l}}{q-1} \tag{2.9}$$

where $b_{r,l}$ is defined as in Lemma 2.1, for $0 \leq l \leq r \leq n-d$. Since the goal is to define a generalized formula for A_{d+i}, by the notation of (2.9), we need $n-l = d+i \implies l = n-d-i$. Since $0 \leq i \leq n-d$, we get $0 \leq l \leq n-d$. Upon substituting $l = n-d-i$ in (2.9), we get

$$\sum_{j=0}^{i} a_j b_{n-d-j,n-d-i} = \frac{A_{d+i}}{q-1} \implies A_{d+i} = (q-1) \sum_{j=0}^{i} a_j b_{n-d-j,n-d-i}. \tag{2.10}$$

For $j > i$, we would have that $n-d-j < n-d-i$, making $b_{n-d-j,n-d-i} = 0$. Next, we expand $b_{n-d-j,n-d-i}$ using Lemma 2.1, where $r = n-d-j$, and $l = n-d-i$. For the sum to be non-empty, we require $n-d-i \leq n-d-j \implies j \leq i$. This holds since $0 \leq j \leq i$. Thus,

$$b_{n-d-j,n-d-i} = \sum_{u=n-d-i}^{n-d-j} \frac{q^{n-d-j-u+1} - 1}{q-1} \binom{n}{u} \binom{u}{n-d-i} (-1)^{u-(n-d-i)}.$$

Let $t = u - (n-d-i)$, then we get $0 \leq t \leq i-j$, and

$$b_{n-d-j,n-d-i} = \sum_{t=0}^{i-j} \frac{q^{i-j-t+1} - 1}{q-1} \binom{n}{t+n-d-i} \binom{n-d-i+t}{t} (-1)^t \tag{2.11}$$

$$\implies b_{n-d-j,n-d-i} = \binom{n}{n-d-i} \sum_{t=0}^{i-j} \frac{q^{i-j-t+1} - 1}{q-1} \binom{d+i}{t} (-1)^t. \tag{2.12}$$

Upon substituting (2.12) in (2.10), we get (2.8). □

Corollary 2.1 (Duursma [2])**.** *If $d^\perp \geq 2$, then $P_C(0) = (q-1)^{-1} \binom{n}{d}^{-1} A_d$, and*

$$\frac{A_{d+1}}{q-1} = \binom{n}{d+1} (P_C(0)(q-d) + P_C'(0)). \tag{2.13}$$

In particular, P_C always has a nonzero positive constant coefficient.

Proof. The proof follows by substituting $i = 0$ and $i = 1$ in (2.8). □

2.2 Second definition

In the second definition of the zeta function, its zeta polynomial is derived from the expression of virtual weight enumerators on the basis of MDS virtual weight enumerators (1.3). We can write the virtual weight enumerator in terms of the MDS virtual weight enumerator of length n and distance d in the following way, analogous to (1.3).

Lemma 2.2 ([4])**.** If F is a virtual weight enumerator of length n and minimum distance d, then there are coefficients $c \in \mathbb{Q}$ and $a_j = a_j(F) \in \mathbb{Q}$ such that

$$F(x, y) = cx^n + a_0 M_{n,d}(x, y) + a_1 M_{n,d+1}(x, y) + \cdots + a_r M_{n,d+r}(x, y) \tag{2.14}$$

for some $r, 0 \leq r \leq n - d$. In fact, $c = 1 - a_0 - \cdots - a_r$.

Definition 2.3. Let $F = A_C$ denote the weight enumerator of a $[n, k, d]_q$−code C. Using the coefficients $a_j = a_j(F)$ of (2.14), define

$$P_C(T) = a_0 + a_1 T + \cdots + a_r T^r. \tag{2.15}$$

This $P_C(T)$ is the *Duursma zeta polynomial of C*.

More generally, if F is a virtual weight enumerator and the coefficients $a_j = a_j(F)$ are as in (2.14), define $P_F(T) = a_0 + a_1 T + \cdots + a_r T^r$. If C is an MDS code of length n and minimum distance d over $\mathbb{F}_q$ then $A_C = M_{n,d}$. This forces $c = 0$, $a_0 = 1$ in (2.14) so $P_C(t) = 1$.

Remark 2.2. For two codes $C, D \subseteq \mathbb{F}_q^n$ that are formally equivalent, i.e., $A_C = A_D$, the zeta polynomials P_C and P_D are equal. Correspondingly, the zeta functions are equivalent, $Z_C = Z_D$. This is because the zeta polynomials are uniquely determined by the weight enumerators (Definition 2.3). The converse does not hold. Two codes may have the same zeta polynomial, but need not have the same weight enumerator.

In the following lemma, we will prove that both the definitions that we gave for the zeta function are equivalent.

Lemma 2.3. The Duursma zeta function of Definition 2.1 is the same as the Duursma zeta function of Definition 2.15.

Proof. In this proof, we will show that the zeta polynomial given by (2.15) satisfies the condition given by (2.1). We start by multiplying $L(T)$ from Definition 2.2 by T^j, for a fixed j. The coefficient of T^{n-d} in this product is the coefficient of T^{n-d-j} in $L(T)$, given by $c_{n-d-j}(x, y)$. Using the MDS definition from Definition 2.2 with $\delta = d + j$, we get that

$$c_{n-(d+j)}(x, y) = \frac{M_{n,d+j}(x, y) - x^n}{(q-1)}.$$

Thus,

$$L(T) \cdot T^j = \cdots + \frac{(M_{n,d+j}(x, y) - x^n)}{(q-1)} \cdot T^{n-d} + \cdots.$$

Upon multiplying both sides by a_j and summing over $j \in \{0, 1, \ldots, r\}$, we get

$$\sum_{j=0}^{r} a_j L(T) \cdot T^j = \cdots + \sum_{j=0}^{r} a_j \frac{M_{n,d+j}(x,y) - x^n}{(q-1)} \cdot T^{n-d} + \cdots .$$

On the left-hand side, we have,

$$L(T) \cdot \sum_{j=0}^{r} a_j T^j = L(T) \cdot P_C(T),$$

by Definition 2.3. And on the right-hand side, we have the coefficient of T^{n-d} given by

$$\sum_{j=0}^{r} a_j \cdot \frac{M_{n,d+j}(x,y) - x^n}{(q-1)} = \frac{1}{q-1}\left[\sum_{j=0}^{r} a_j \cdot M_{n,d+j}(x,y) - (a_0 + \cdots + a_r)x^n\right]. \tag{2.16}$$

From Definition 2.3 and Lemma 2.2 we have, $\sum_{j=0}^{r} a_j M_{n,d+j} = A_C(x,y) - cx^n$, and $a_0 + \cdots + a_r = 1 - c$. We substitute this back into (2.16) to get

$$\frac{1}{q-1}\left[A_C(x,y) - cx^n - x^n + cx^n\right] = \frac{A_C(x,y) - x^n}{q-1}.$$

Thus,

$$L(T).P_C(T) = \cdots + \frac{A_C(x,y) - x^n}{q-1} \cdot T^{n-d} + \cdots .$$

Since the zeta polynomial is unique (Proposition 2.2), any polynomial $P_C(T)$ that makes the coefficient of T^{n-d} equal to $\frac{A_C(x,y)-x^n}{q-1}$ must be the zeta polynomial. The polynomial $P_C(T) = \sum_{j=0}^{r} a_j T^j$ from Definition 2.3 has this property. Therefore, the two definitions are equivalent. □

2.3 Third definition

The third definition of the zeta function is inspired by the definition of the zeta function of algebraic curves as a generating function for the number of effective divisors (Definition 1.11). We will define the zeta function of a linear code as the generating function for the normalized binomial moments of the code. This definition originated in [14].

For a linear code C, of length n over the finite field $\mathbb{F}_q$, let C_S be the subcode of C of words with support on $S \subset \{1, 2, \ldots, n\}$. Let k_S denote the dimension of C_S.

$$k_S = \begin{cases} 0, & \text{for } 0 \leq |S| < d. \\ k - (n - |S|), & \text{for } n - d^{\perp} < |S| \leq n. \end{cases}$$

Since the dimension of C_S is k_S, we have that $|C_S| = q^{k_S}$. The set of non-zero codewords in $|C_S|$ is $(C_S \setminus \{0\})/\mathbb{F}_q^{\times}$, and the size of C_{k_S} is $q^{k_S} - 1$, and the size of $\mathbb{F}_q^{\times}$ is $q - 1$. Thus, the number of nonzero words in C_S counted up to multiplication by scalars is given by $(q^{k_S} - 1)/(q - 1)$.

Definition 2.4. Let

$$B_i^1 = \sum_{|S|=i} \frac{q^{k_S} - 1}{q - 1}.$$

These B_i are called the *binomial moments* of the code. And we get

$$B_i^1 = \begin{cases} 0, & \text{for } 0 \le i < d. \\ k - (n-i), & \text{for } n - d^\perp < i \le n. \end{cases}$$

Definition 2.5. Let $b_i = B_{d+i}^1 / \binom{n}{d+i}$ be the *normalized binomial moment*. We can extend the definition of b_i to all $i \in \mathbb{Z}$, by setting

$$b_i = \begin{cases} 0, & \text{for } i < 0. \\ (q^{i+d+k-n} - 1)/(q-1), & \text{for } i > n - d - d^\perp. \end{cases}$$

Now we can give the third definition of the zeta function of a linear code in the Hamming metric.

Definition 2.6 (Third definition of the zeta function)**.** Define the zeta function as the generating function

$$Z_C(T) = \sum_{i \ge 0} b_i T^i. \tag{2.17}$$

For $i \notin \{0, 1, \ldots, n - d - d^\perp + 2\}$, we have $b_i - (q+1)b_{i-1} + qb_{i-2} = 0$.

Let

$$p_i = b_i - (q+1)b_{i-1} + qb_{i-2}. \tag{2.18}$$

Upon multiplying both sides by T^i, and summing over all $i \ge 2$, we get,

$$\sum_{i \ge 2} b_i T^i - \sum_{i \ge 2} (q+1) b_{i-1} T^i + \sum_{i \ge 2} q b_{i-2} T^i = \sum_{i \ge 2} p_i T^i. \tag{2.19}$$

From (2.17) and (2.19) we get,

$$Z_C(T) - b_0 - b_1 T - (q+1)T(Z_C(T) - b_0) + qT^2 Z_C(T) = \sum_{i \ge 2} p_i T^i \tag{2.20}$$

$$\implies Z_C(T)(1-T)(1-qT) - b_0 + T[b_0(1+q) - b_1] = \sum_{i=2}^{n-d-d^\perp+2} p_i T^i. \tag{2.21}$$

From (2.18), we have that $p_0 = b_0$, and $p_1 = b_1 - (q+1)b_0$. Substituting this in (2.21) gives us

$$Z_C(T)(1-T)(1-qT) = p_0 + p_1 T + \sum_{i=2}^{n-d-d^\perp+2} p_i T^i = \sum_{i=0}^{\infty} p_i T^i = P_C(T),$$

since by definition, for higher i values, p_i will be zero. Thus, we can write the zeta function as the rational function

$$Z_C(T) = \frac{P_C(T)}{(1-T)(1-qT)},$$

where $P_C(T) = p_0 + p_1 T + \cdots + p_{n+2-d-d^\perp} T^{n+2-d-d^\perp}$.

For the one variable zeta function, we have the following relation from [14], analogous to the relation

for the one-variable zeta function of algebraic curves (1.13),

$$(q-1)T^{k+d-n}Z_C(T) = \sum_{j=0}^{n}\left(\frac{1}{\binom{n}{j}}\sum_{|S|=j} q^{k_S} - 1\right)T^{k+j-n} + \sum_{j>n}(q^{j+k-n}-1)T^{j+k-n}.$$

We define the two-variable zeta function for linear codes by replacing q with the variable u. Then

$$Z_C(T,u) = \frac{P_C(T,u)}{(1-T)(1-uT)}$$

is a rational function with $\deg_T P_C(T,u) = n+2-d-d^\perp$, and we get the following relation analogous to the two-variable Pellikaan zeta function of algebraic curves (1.20),

$$(u-1)T^{k+d-n}Z_C(T,u) = \sum_{j=0}^{n}\left(\frac{1}{\binom{n}{j}}\sum_{|S|=j} u^{k_S} - 1\right)T^{k+j-n} + \sum_{j>n}(u^{j+k-n}-1)T^{j+k-n}.$$

Lemma 2.4 ([4], Lemma 110)**.** The Duursma zeta function of Definition 2.1 is the same as the Duursma zeta function of Definition 2.6.

2.4 Puncturing and shortening

While the definitions of the zeta function for a linear code are well-defined, they may not always be convenient for explicit computation. A different way to approach the zeta function is by considering how the code can be obtained from a longer code by the operations of puncturing and shortening.

For a linear code C of length n, choose a coordinate position $i \in \{1,\dots,n\}$. Then the *punctured code* $\mathcal{P}_i(C)$ is obtained by deleting the i-th coordinate from every codeword:

$$\mathcal{P}_i(C) := \{(c_1,\dots,c_{i-1},c_{i+1},\dots,c_n) \mid (c_1,\dots,c_n) \in C\}.$$

If C has dimension k and no non-zero codeword has a zero in the i-th coordinate, i.e., the i-th coordinate is not always zero, then $\dim \mathcal{P}_i(C) = k$. Otherwise, $\dim \mathcal{P}_i(C) = k-1$. The punctured code has length $n-1$. And for a fixed coordinate i, the *shortened code* $\mathcal{S}_i(C)$ is defined as

$$\mathcal{S}_i(C) = \{(c_1,\dots,c_{i-1},c_{i+1},\dots,c_n) \mid (c_1,\dots,c_n) \in C \text{ and } c_i = 0\}.$$

The shortened code has length $n-1$. Its dimension is $\dim \mathcal{S}_i(C) = \dim C - 1$ if there exists at least one codeword with $c_i = 1$, otherwise the shortening has no effect on the dimension and $\dim \mathcal{S}_i(C) = \dim C$.

As we can see, the effect of these operations depends on the choice of the coordinates that we choose to delete or fix. To avoid this dependency, we consider instead the average of the n weight enumerators obtained by puncturing or shortening at all n coordinates individually. This average depends only on the weight enumerator of the parent code. A codeword of weight i (or the term $x^{n-i}y^i$ from the weight enumerator polynomial) contributes with $n-i$ words of weight i and i words of weight $i-1$ to the n punctured codes (or with $(n-i)x^{n-1-i}y^i + ix^{n-i}y^{i-1}$ to the sum of their weight enumerator polynomials). And with $n-i$ words of weight i to the n shortened codes (or with $(n-i)x^{n-1-i}y^i$ to the sum of their weight enumerator

polynomials). Then we get the average of the n punctured weight enumerators to be

$$\bar{A}_{\text{punc}} = \frac{1}{n}\sum_{i=0}^{n} A_i[(n-i)x^{n-1-i}y^i + ix^{n-i}y^{i-1}],$$

and the effect of “puncturing and averaging" may be described by the operators

$$\mathcal{P} = \left(\frac{\partial}{\partial_x} + \frac{\partial}{\partial_y}\right)/n.$$

The average of the n shortened weight enumerators comes out to be

$$\bar{A}_{\text{short}} = \frac{1}{n}\sum_{i=0}^{n} A_i(n-i)x^{n-1-i}y^i,$$

and the effect of “shortening and averaging" may be described by the operators

$$\mathcal{S} = \left(\frac{\partial}{\partial_x}\right)/n.$$

As arithmetic manipulations, $\mathcal{P}(A)$ and $\mathcal{S}(A)$ depend on the weight enumerator A but not on the code that realizes the weight enumerator. We will refer to $\mathcal{P}(A_C)$ and $\mathcal{S}(A_C)$ as the punctured and shortened codes, respectively.

Example 2.1. Let us consider the weight enumerator polynomial of the ternary Golay code C with parameters $[12, 6, 6]$, given by

$$A_C(x, y) = \frac{1}{7}(M_{12,6} + 3M_{12,7} + 3M_{12,8}).$$

To obtain the weight enumerator of the punctured code, we apply the operator $\mathcal{P}$ and obtain

$$\mathcal{P}(A_C(x, y)) = \frac{1}{7}(M_{11,5} + 3M_{11,6} + 3M_{11,7}).$$

Similarly, for the shortened code we obtain

$$\mathcal{S}(A_C(x, y)) = \frac{1}{7}(M_{11,6} + 3M_{11,7} + 3M_{11,8}).$$

In both cases, the zeta polynomial has not changed, since by Definition 2.3, we know that the coefficients of the zeta polynomial coincide with the coefficients of (1.3), and because puncturing or shortening an MDS code yields again an MDS code. To find the weight enumerators, we use (2.1) with the zeta polynomial of the ternary Golay code but with $n = 11$ instead of $n = 12$, to get

$$\cdots + (66x^5y^6 + 55x^2y^9)T^5 + (66x^6y^5 + 66x^5y^6 + 165x^3y^8 + 55x^2y^9 + 12y^{11})T^6 + \cdots.$$

We get the weight enumerators of the shortened and the punctured Golay code from the coefficients of T^5 and T^6, respectively. This is also a good example to highlight the observation that we made in Remark 2.2. We can observe here that the shortened and punctured codes have the same zeta polynomial but different weight enumerators.

We can generalize this property.

Proposition 2.4 ([2])**.** *The zeta function of a linear code is invariant under the operations $\mathcal{P}$ (puncturing and averaging) and $\mathcal{S}$ (shortening and averaging).*

But the weight enumerator of the linear code is not invariant under the operations $\mathcal{P}$ and $\mathcal{S}$. We require the left-hand side and right-hand side of (2.1) to match. Thus, a better relation and transformation between the zeta function and weight enumerator arises if we normalize the weight enumerator in such a way that it becomes invariant under the operations $\mathcal{P}$ and $\mathcal{S}$.

Definition 2.7. Associate to the weight enumerator $A_C(x, y)$ the polynomial

$$a(t) = \frac{1}{q-1} \sum_{i=d}^{n} A_i / \binom{n}{i} t^{i-d}.$$

Proposition 2.5. *For the weight enumerator $A_C(x, y)$ with associated polynomial $a(t)$ the associated polynomials $a_{\mathcal{P}}(t)$ and $a_{\mathcal{S}}(t)$, of the punctured enumerator $\mathcal{P}(A_C)$ and the shortened enumerator $\mathcal{S}(A_C)$, respectively, are*

$$a_{\mathcal{P}}(t) = (1+t)a(t) - A_n t^{n-d+1}, \qquad a_{\mathcal{S}}(t) = a(t) - A_n t^{n-d}. \tag{2.22}$$

Proof. Appendix B [2]. □

Theorem 2.1. *The expression $a(t)(1+t)^d \pmod{t^{n-d+1}}$ is invariant under $\mathcal{P}$, and $\mathcal{S}$.*

Proof. With parameters $n_{\mathcal{P}} = n - 1$ and $d_{\mathcal{P}} = d - 1$ for the punctured code and $n_{\mathcal{S}} = n - 1$ and $d_{\mathcal{S}} = d$ for the shortened code, we need to prove that

$$a_{\mathcal{P}}(t)(1+t)^{d_{\mathcal{P}}} \equiv a(t)(1+t)^d \pmod{t^{n_{\mathcal{P}}-d_{\mathcal{P}}+1}}, \tag{2.23}$$

$$a_{\mathcal{S}}(t)(1+t)^{d_{\mathcal{S}}} \equiv a(t)(1+t)^d \pmod{t^{n_{\mathcal{S}}-d_{\mathcal{S}}+1}}. \tag{2.24}$$

It follows by substituting (2.22) in (2.23) and (2.24). □

Next, we determine the relation between the zeta polynomial $P_C(T)$ and the normalized weight enumerator $a(t)$ of a code.

Theorem 2.2. *For a given weight enumerator $A_C(x, y)$, the zeta polynomial $P_C(T)$ and the normalized weight enumerator $a(t)$ are related by*

$$\frac{P_C(T)}{(1-T)(1-qT)}(1-T)^{d+1} \equiv a\left(\frac{T}{1-T}\right) \pmod{T^{n-d+1}}. \tag{2.25}$$

Proof. Let $A = (\alpha\mathcal{P} + \beta\mathcal{S})A'$ be a weight enumerator of length n and minimum distance d, where A' has length $n+1$, and minimum distance $d+1$. When we form $a(t)$ from A, it is normalized with respect to the minimum distance d. The terms $\mathcal{P}(A')$ and $\mathcal{S}(A')$ correspond to codes of minimum distance d and $d+1$, respectively. The normalized polynomial of $\mathcal{P}(A')$ is $(1+t)a'(t)$, ignoring the higher power terms. And the normalized polynomial of $\mathcal{S}(A')$ is $a'(t)$, but we need to express it with respect to the minimum distance d, since A uses d, so we multiply it by t. Thus, upon applying Proposition 2.5, we get

$$a(t) \equiv (\alpha(1+t) + \beta t)a'(t) \pmod{t^{n-d+1}}. \tag{2.26}$$

After a change of variable $T = t/(1+t)$, and $t = T/(1-T)$, (2.26) becomes

$$a\left(\frac{T}{1-T}\right) \equiv \frac{(\alpha+\beta T)}{(1-T)} a'\left(\frac{T}{1-T}\right) \pmod{T^{n-d+1}}. \tag{2.27}$$

Let M be the MDS code of length $n+r$ and minimum distance $d+r$ with associated polynomial $m(t)$. From (1.3), we can write

$$A_C(x,y) = (a_0\mathcal{P}^r + a_1\mathcal{P}^{r-1}\mathcal{S} + \cdots + a_r\mathcal{S}^r)M.$$

The coefficients $a_0, \ldots, a_r$ are exactly the coefficients of the zeta polynomial from Definition 2.3. Every time we apply $\mathcal{P}$ or $\mathcal{S}$, we reduce the length by 1 and get a term from (2.27). After r steps, the product of the $(1-T)$ denominators accumulates to $(1-T)^r$. The numerator becomes $P_C(T)$. Hence we get

$$a\left(\frac{T}{1-T}\right) \equiv \frac{P_C(T)}{(1-T)^r} m\left(\frac{T}{1-T}\right) \pmod{T^{n-d+1}}. \tag{2.28}$$

We still need to determine $m\left(\frac{T}{1-T}\right)$. After puncturing M, $d+r-1$ times, we obtain the weight enumerator of the full space of length $v = n+r-(d+r-1) = n+1-d$. The full space of length v has weight enumerator $A_C(x,y) = (x+(q-1)y)^v$ and $a(t) = \sum_{i=1}^{v}(q-1)^{i-1}t^{i-1}$. Its minimum distance is one, and the invariant expression in Theorem 2.1 becomes

$$a(t)(1+t) \equiv \frac{1+t}{1-(q-1)t} \pmod{t^v}.$$

With Theorem 2.1, the unpunctured MDS code M gives the same expression,

$$m(t)(1+t)^{d+r} \equiv \frac{1+t}{1-(q-1)t} \pmod{t^{n-d+1}}. \tag{2.29}$$

After a change of variable, this becomes

$$m\left(\frac{T}{1-T}\right) \equiv \frac{(1-T)^{d+r+1}}{(1-T)(1-qT)} \pmod{T^{n-d+1}}. \tag{2.30}$$

Upon combining (2.28) and (2.30), we get (2.25). □

In Theorem 4 of [2], Duursma uses Theorem 2.1 to obtain the zeta polynomials of random divisible self-dual weight enumerators explicitly. An interesting observation to be made about these zeta polynomials is that they all have roots of the same modulus, which is $1/\sqrt{q}$. We will see why this might be of importance in the next section.

2.5 The Riemann hypothesis analogue

In this section, we discuss the analogies between the Duursma zeta function of codes in the Hamming metric and the Hasse-Weil zeta function of algebraic curves, extending to parallels between the Riemann hypothesis analogue of codes and algebraic curves. We will particularly focus on the Riemann hypothesis analogue of self-dual linear codes in the Hamming metric since it exhibits stronger analogies to the Riemann hypothesis of algebraic curves.

The Riemann hypothesis analogue of both codes and algebraic curves focuses on the reciprocal of zeros of the zeta polynomial. In the case of algebraic curves, the reciprocals of zeros are crucial in obtaining upper

bounds on the relative number of rational points of an algebraic curve. And for self-dual linear codes, they are used to derive upper bounds for the relative minimum distance of a self-dual linear code.

2.5.1 Motivation

The Duursma zeta function of codes has a finite number of complex zeros, which are calculated using the zeta polynomial in the numerator. For codes that satisfy certain properties, they lie on a circle of constant radius. This suggests that there may exist an analogue of the Riemann hypothesis for codes. Using this, we can obtain improved upper bounds on the weight distribution, which in turn gives bounds on code parameters. In particular, the knowledge of zeros of $Z(T)$ might be very useful in understanding the possible values of the minimum distance.

Let C be a code that is not MDS. And let $\delta_1, \delta_2, \ldots, \delta_r$ denote the zeros, counted according to multiplicity, of the Duursma zeta polynomial $P_C(t)$ of a linear code C, then

$$P_C(T) = a_0 + a_1 T + \cdots + a_r T^r = a_r \left(T^r + \frac{a_{r-1}}{a_r} t^{r-1} + \cdots + \frac{a_1}{a_r} t + \frac{a_0}{a_r} \right) = a_r \cdot Q(T),$$

where $Q(T)$ is monic. And since $a_r \neq 0$, the zeros of $Q(T)$ are the same as the zeros of $P_C(T)$. So we can factor $Q(T)$ as

$$Q(T) = \prod_{i=1}^{r} (T - \delta_i), \tag{2.31}$$

where δ_i's are the zeros. Thus,

$$P_C(T) = a_r \prod_{i=1}^{r} (T - \delta_i).$$

Upon taking the logarithmic derivative of this, we obtain

$$\frac{\mathrm{d}}{\mathrm{dT}}(\log P_C(T)) = \frac{\mathrm{d}}{\mathrm{dT}}(\log a_r) + \frac{\mathrm{d}}{\mathrm{dT}}(\log(\prod_{i=1}^{r}(T - \delta_i))) = \sum_{i=1}^{r} \frac{1}{(T - \delta_i)}.$$

This implies that

$$\frac{P'_C(T)}{P_C(T)} = \sum_{i=1}^{r} \frac{1}{(T - \delta_i)}.$$

We have the following proposition by Duursma that explicitly gives the relation between the reciprocal zeros of the zeta polynomial of a code and its minimum distance.

Proposition 2.6 (Duursma [2])**.** *If $A_0, \ldots, A_n$ denotes the weight distribution of C, then*

$$d = q - \sum_i \delta_i^{-1} - \frac{A_{d+1}}{A_d} \frac{d+1}{n-d}. \tag{2.32}$$

In particular,

$$d \leq q - \sum_i \delta_i^{-1}. \tag{2.33}$$

Proof. From Corollary 2.1, we have

$$\frac{A_{d+1}}{q-1} = \binom{n}{d+1} (P_C(0)(q-d) + P'(0)) \implies \frac{1}{q-1} = \binom{n}{d+1} \frac{a_0(q-d) + a_1}{A_{d+1}} \tag{2.34}$$

and

$$P_C(0) = (q-1)^{-1}\binom{n}{d}^{-1} A_d \implies \frac{1}{q-1} = \frac{a_0}{A_d}\binom{n}{d}. \tag{2.35}$$

Upon equating $(1/(q-1))$ from Equations (2.34) and (2.35), we get

$$\binom{n}{d+1}\frac{a_0(q-d)+a_1}{A_{d+1}} = \frac{a_0}{A_d}\binom{n}{d} \implies d = q + \frac{a_1}{a_0} - \frac{A_{d+1}}{A_d}\frac{(d+1)}{(n-d)}. \tag{2.36}$$

We factor $P_C(t)$ with respect to the coefficient a_0 to obtain

$$P_C(t) = a_0\left(1 + \frac{a_1}{a_0}t + \frac{a_2}{a_0}t^2 + \cdots + \frac{a_r}{a_0}t^r\right) = a_0 \cdot R(t). \tag{2.37}$$

Since $R(t)$ is a polynomial of degree r with constant term 1, its roots δ_i satisfy

$$R(t) = \prod_{i=1}^{r}\left(1 - \frac{t}{\delta_i}\right) = \left(1 - \frac{t}{\delta_1}\right)\left(1 - \frac{t}{\delta_2}\right)\cdots\left(1 - \frac{t}{\delta_r}\right) = 1 - \left(\sum_{i=1}^{r}\frac{1}{\delta_i}\right)t + \cdots. \tag{2.38}$$

Upon comparing the coefficients of t in Equations (2.37) and (2.38), we get

$$\frac{a_1}{a_0} = -\sum_{i=1}^{r}\frac{1}{\delta_i}. \tag{2.39}$$

We substitute this in (2.36) to get the required relation given by (2.32). The second statement follows from the first since $\frac{A_{d+1}}{A_d} \geq 0$, and $\frac{d+1}{n-d}$ is positive because $d \geq 2$ and $d < n$ for a non-trivial code. □

A consequence of Duursma's proposition can be seen for a b-divisible code with $b \geq 2$. If C is b-divisible, then all the weights of C are multiples of b. The minimum distance d is a multiple of b if $b \geq 2$, but then $d+1$ is not a multiple of b because consecutive integers cannot both be divisible by $b \geq 2$. Thus, $d+1$ is not the weight of any codeword in a b-divisible code, so $A_{d+1} = 0$, and

$$d = q - \sum_i \delta_i^{-1}.$$

It is clear that the reciprocal zeros of the zeta polynomial are important.

Remark 2.3. The duality of the zeta function for a linear code is given by (2.4). For self-dual linear codes, we had that $Z_{C^\perp}(T) = Z_C(T)$ and $g = g^\perp$, so (2.4) becomes

$$Z_C(T) = Z_C\left(\frac{1}{qT}\right)q^{g-1}T^{2g-2}. \tag{2.40}$$

The relation (2.40) matches the functional equation under duality of the zeta function of algebraic curves (1.12). This is why the analogies between the zeta function of an algebraic curve and the zeta function of a linear code are closer for self-dual linear codes. For self-dual linear codes, the zeta polynomial satisfies (2.40). Therefore, the reciprocal zeros different from $\pm\sqrt{q}$ appear in pairs $(\alpha, 1/q\alpha)$. For the zeta functions of algebraic curves, it is known that all reciprocal zeros have the same absolute value $\sqrt{q}$ (Theorem 1.5). The Riemann hypothesis of algebraic curves over a finite field is crucial in obtaining upper bounds for the number of rational points on an algebraic curve (Theorem 1.6). To obtain upper bounds on the minimum distance of a self-dual linear code in a similar way, Duursma [9] gave a Riemann hypothesis for self-dual

linear codes.

2.5.2 The Riemann hypothesis analogue for self-dual codes

Definition 2.8 ([9])**.** A self-dual linear code satisfies the *analogue of the Riemann hypothesis* (has the RH property) if all the reciprocal zeros of its zeta polynomial have absolute value $\sqrt{q}$.

Duursma's paper [9] explores an analogue of the Riemann hypothesis for some self-dual linear codes whose weight enumerators are explicitly known. They find many examples for which it holds. But not all self-dual linear codes satisfy the RH property. We will go into detail on this in the next section. Before that, we have the following result illustrating that, for large q, the Riemann hypothesis is often false. It shows that constructing codes that violate the Riemann hypothesis is not hard.

Corollary 2.2. *Let C be an $[n, k, d]_q$ code over $\mathbb{F}_q$ with $A_{d+1} = 0$, $q > n^2$, $2 \leq d$, and $d + d^\perp < n + 2$. If $n > 3$, then the Duursma zeta polynomial is not a constant and does not satisfy the Riemann hypothesis.*

Proof. If $n > 3$ then using $d \geq 2$ and the existence criteria for $P_C(T)$, i.e., $d^\perp \geq 2$, we get a bound on the $\deg P_C(T) = n + 2 - d - d^\perp$ to be $n + 2 - d - d^\perp \geq 4 + 2 - 2 - 2 = 2 > 1$.This implies that $P_C(T)$ is not a constant. We prove the second part by contradiction. Let $\delta_1, \ldots, \delta_r$ be the zeros of $P_C(T)$. Let $n > 3$ and suppose $P_C(T)$ satisfies the Riemann hypothesis, i.e., $|\delta_i^{-1}| = \sqrt{q}$ for $i = 1, \ldots, r$, and $q > n^2$. Then we have,

$$\sum_{i=1}^{r} |\delta_i^{-1}| = \sum_{i=1}^{r} \sqrt{q} = r\sqrt{q}. \tag{2.41}$$

Using the triangle inequality, we get

$$|\sum_{i=1}^{r} \delta_i^{-1}| \leq \sum_{i=1}^{r} |\delta_i^{-1}|. \tag{2.42}$$

Upon combining Equations (2.41) and (2.42), we get

$$|\sum_{i=1}^{r} \delta_i^{-1}| \leq r\sqrt{q}. \tag{2.43}$$

And upon substituting $A_{d+1} = 0$ in (2.32), we get

$$d = q - \sum_{i=1}^{r} \delta_i^{-1} \implies q - d = \sum_{i=1}^{r} \delta_i^{-1} \implies |q - d| = |\sum_{i=1}^{r} \delta_i^{-1}|. \tag{2.44}$$

We combine Equations (2.43) and (2.44) to get

$$|q - d| \leq r\sqrt{q} = (n + 2 - d - d^\perp)\sqrt{q}. \tag{2.45}$$

Since $q > n^2$ and $d \leq n$, we have $q - d > 0$. Thus, $|q - d| = q - d$, and we can simplify (2.45) to $q - d \leq r\sqrt{q} = (n + 2 - d - d^\perp)\sqrt{q}$. Since $d^\perp \geq 2$ and $d \geq 2$: $r = n + 2 - d - d^\perp \leq n - 2$. Thus, $q - d \leq (n - 2)\sqrt{q}$. And since $q - d > q - n$, $q - n < (n - 2)\sqrt{q}$. Upon dividing both sides by $\sqrt{q}$, we get $\sqrt{q} - \frac{n}{\sqrt{q}} < n - 2$, and since $q > n^2 \implies \sqrt{q} > n$, we have $\frac{n}{\sqrt{q}} < 1$. Thus, $\sqrt{q} - 1 < n - 2 \implies \sqrt{q} < n - 1$. This contradicts $\sqrt{q} > n$. □

Since the Duursma zeta function of a code only depends on the code via its weight enumerator polynomial, we have that for any virtual weight enumerator polynomial $F(x, y)$, there is an associated zeta function Z_F

and zeta polynomial P_F. If we define $F^\perp$ by $F^\perp = F \circ \sigma$, where

$$\sigma = \frac{1}{\sqrt{q}} \begin{pmatrix} 1 & q-1 \\ 1 & -1 \end{pmatrix},$$

then there are functional equations relating Z and $Z_{F^\perp}$, and P and $P_{F^\perp}$. Even though F does not depend on q, it is important to note that $F^\perp$ and $Z_{F^\perp}$ do.

The following proposition defines them formally.

Proposition 2.7 ([4], Chapter 4, Proposition 111)**.** *For a virtually self-dual weight enumerator F satisfying*

$$F(x,y) = a_0 M_{n,d}(x,y) + a_1 M_{n,d+1}(x,y) + \cdots + a_r M_{n,d+r}(x,y)$$

and for any q, the zeta function Z_F satisfies the functional equation

$$Z^\perp(T) t^{1-g^\perp} = Z\left(\frac{1}{qT}\right)\left(\frac{1}{qT}\right)^{1-g}.$$

Analogously, the zeta polynomial P_F satisfies the functional equation

$$P^\perp(T) = P\left(\frac{1}{qT}\right) q^g T^{g+g^\perp},$$

where $g = n/2 + 1 - d$ and $g^\perp = n/2 + 1 - d^\perp$.

It is important to note that both P and $P^\perp$ are polynomials of degree $n + 2 - d - d^\perp = g + g^\perp$ and g is the genus if $F = A_C$ is an actual weight enumerator. A detailed proof of this proposition can be found in [4]. Using this information, we can generalize the notion of the Riemann hypothesis of codes to define the Riemann hypothesis of self-dual virtual weight enumerators.

Definition 2.9. We say that the *zeta function* Z_F of a virtual self-dual weight enumerator F satisfies the *Riemann hypothesis* (has the RH property) if all the reciprocal zeros of its zeta polynomial have absolute value $\sqrt{q}$.

2.5.3 Duursma's conjecture

In [2], Duursma derives an upper bound for the relative minimum distance of a self-dual linear code under the assumption that its zeta function also forms a positive Weil system [15], like the zeta function of an algebraic curve (1.26). We will discuss this in detail in this section.

Recall that for self-dual linear codes, we have that $P_C(T) = P_{C^\perp}(T)$ and it follows that $P_C(\alpha) = 0$ if and only if $P_C(1/q\alpha) = 0$. And $P_C(T)$ allows a factorization (for self-dual linear codes)

$$P_C(T) = a_0 \prod_{i=1}^{g} (1 + b_i T + qT^2). \tag{2.46}$$

Since the theory of the zeta functions of linear codes is moving analogously to the zeta functions of algebraic curves, the natural question that arises now is if the zeta functions of self-dual linear codes (2.46) will form a positive Weil system like the zeta functions of algebraic curves (1.26) do.

If the zeta function of self-dual linear codes formed a positive Weil system, then we could use the "explicit formulas" method that we did for algebraic curves, to obtain a bound similar to the Drinfeld-Vladtus bound (Theorem 1.7).

Let us explicitly understand the correspondence between algebraic curves and self-dual linear codes. The degree of the zeta polynomial is given by

$$2g \quad \text{and} \quad n+2-2d$$

for algebraic curves and self-dual linear codes, respectively. From (1.25), we know that the number of rational points depends on the sum of the reciprocal zeros of the zeta polynomial $P_{\mathcal{X}}(T)$, i.e., $\sum_{i=1}^{2g} \alpha_i$. Let $a' = -\sum_{i=1}^{2g} \alpha_i$. Then, from (1.25), we get

$$N = q + 1 + a'. \tag{2.47}$$

And in the case of self-dual linear codes, we can rearrange (2.13) in terms of the reciprocal zeros of $P_C(T)$, i.e., $\sum_i \delta_i^{-1} = -a'$, to get

$$\frac{A_{d+1}}{q-1} = a_0 \binom{n}{d+1}(q+1+a') - a_0 \binom{n}{d+1}(d+1),$$

and it follows from (2.36) and (2.39) that $d = q + a' - \frac{A_{d+1}(d+1)}{A_d(n-d)}$ which is equivalent to

$$d + 1 + \frac{\alpha_1}{\alpha_0} = q + 1 + a', \tag{2.48}$$

where $\alpha_i = \frac{A_{d+i}}{\binom{n}{d+i}}(q-1)$. From (2.47) and (2.48), we get the analogue between algebraic curves and self-dual linear codes in terms of the sum of their reciprocal zeros $-a'$ to be

$$N = q + 1 + a' \quad \text{and} \quad q + 1 + a' = d + 1 + \frac{\alpha_1}{\alpha_0}.$$

In each case, we have an optimization problem [2]

Maximize the number of points N for a given g

Maximize the minimum distance d for a given length n

with known upper bounds of the form

$$N \leq \gamma_q + c_q \quad \text{and} \quad d \leq \alpha_q n + \beta_q$$

for an algebraic curve and for a self-dual linear code, respectively. In general, for self-dual linear codes, $\alpha_1/\alpha_0 \geq 0$ is small. It is zero for self-dual binary and ternary codes. Thus, the analogy gives us a correspondence for the parameters

$$(N, g) = \left(d + 1, \frac{n+2-2d}{2}\right).$$

We can plug this into the Drinfeld-Vladtus bound (Theorem 1.7) to get the following bound for the minimum

relative distance.

Theorem 2.3 ([2] Theorem 6)**.** *For an infinite family of self-dual linear codes of increasing length n that are defined over a field of q elements and that have a zeta function whose reciprocal zeros define a positive Weil system (Definition 1.14),*

$$\limsup_{n\to\infty} d/n \le \delta_{EF} = \frac{1}{2} - 1/2\sqrt{q}.$$

In particular,

$$\begin{aligned} q = 2: &\quad \limsup_{n\to\infty} d/n \le \frac{1}{2} - 1/(2\sqrt{2}) = 0.146\ldots, \\ q = 3: &\quad \limsup_{n\to\infty} d/n \le \frac{1}{2} - 1/(2\sqrt{3}) = 0.213\ldots, \\ q = 4: &\quad \limsup_{n\to\infty} d/n \le \frac{1}{2} - 1/(2\sqrt{4}) = 0.25. \end{aligned}$$

Duursma [2] compares these bounds with all the best-known upper bounds, and in each case, the bounds have been improved by Theorem 2.3.

Keeping the relative minimum distance bound in mind, we are particularly interested in verifying the Riemann hypothesis for extremal self-dual linear codes (Definition 1.10) that are uniquely determined by their parameters d and n. If there exist any codes whose zeta functions could satisfy the Riemann hypothesis, they are most likely to be extremal self-dual linear codes.

In [12], Duursma explicitly derives the formulas for the weight enumerators of all extremal self-dual linear codes of type I-IV. And then they transform these weight enumerators into zeta functions. They prove that the Riemann hypothesis analogue is true for extremal self-dual type IV codes, and conjecture that it may hold for extremal self-dual linear codes of type I, II, and III. And since extremal self-dual linear codes have the largest possible d for a given n, this bound would apply to all self-dual linear codes. In other words, if every extremal self-dual linear code satisfies the Riemann hypothesis analogue, then the conjectural δ_{EF} is a universal upper bound for all self-dual linear codes.

Chapter 3

Zeta functions of rank-metric codes

The notations and theory in this chapter follow those of [3] closely. Analogous to the Hamming case, we will give three definitions of the zeta function of codes in the rank-metric. Following that, we will discuss the Riemann hypothesis analogue for rank-metric codes, which is an open problem.

3.1 First definition

In this section, we will define the zeta function of a rank-metric code in terms of its binomial moments, analogous to the Hamming metric case treated in Section 2.3. To do this, we require the notion of a shortened code. It is defined in the following way.

Definition 3.1. Let $U \subseteq \mathbb{F}_q^n$ be a subspace of dimension u. The *shortened subcode of* C with respect to U is

$$C_U := \{X \in C : U \leq X^\perp\}.$$

The *strict shortening of* C by U is

$$\hat{C}_U := \{X \in C : U = X^\perp\}.$$

Using the rank-nullity theorem, we know that for a matrix $X \in \hat{C}_U$, $\dim(\mathbb{F}_q^n) = \mathrm{rk(X)} + \dim(X^\perp) \implies \mathrm{rk}(X) = n - u$.

Definition 3.2. For $u \geq 0$, the uth binomial moment of C is defined by

$$B_u(C) := \sum_{dim(U)=n-u} (|C_U| - 1)$$

Lemma 3.1 ([20], Lemma 28)**.** Let U be a subspace of $\mathbb{F}_q^n$ of dimension $n - u$. Then

$$|C_U| = \frac{|C||C^\perp_{U^\perp}|}{q^{m(n-u)}}.$$

Lemma 3.2. Let C have dimension k and minimum rank distance d, and $C^\perp$ have minimum distance $d^\perp$. Then

$$B_u(C) = \begin{cases} 0, & \text{if } u < d. \\ q^{k-m(n-u)} - 1 \begin{bmatrix} n \\ u \end{bmatrix}, & \text{if } u > n - d^\perp. \end{cases}$$

Proof. By the rank-nullity theorem, we have that $\dim(X^\perp) = n - \mathrm{rk}(X) \leq n - d$ since $\mathrm{rk}(X) \geq d$. Thus, any $X \in C$ satisfies $\mathrm{rk}(X^\perp) \leq n - d$, so if U has dimension $n - u > n - d$, then by definition of C_U, this would imply that $\mathrm{rk}(X^\perp) > n - d$, which is not possible. Thus, for $n - u > n - d$, $C_U = \{0\}$. Similarly, $C^\perp_{U^\perp} = \{0\}$ if $\dim(U^\perp) = u > n - d^\perp$ since every element $X \in C^\perp$ has rank atleast $d^\perp$. From Lemma 3.1, we get that if $u < d \implies n - u > n - d$, then $|C_U| = 1$. And if $u > n - d^\perp$, then $|C^\perp_{U^\perp}| = 1$, and $|C| = q^k$, so $|C_U| = q^{k-m(n-u)}$. From Definition 3.2, we have that for $u < d$, $B_U(C) = 0$. And when $u > n - d^\perp$, $B_U(C) = (q^{k-m(n-u)} - 1) \begin{bmatrix} n \\ u \end{bmatrix}$, since $\begin{bmatrix} n \\ u \end{bmatrix}$ counts the number of subspaces of dimension u. □

Using Lemma 3.1 and the singleton bound, we can deduce that if U has dimension u, then

$$|C_U| \leq \frac{|C|}{q^{mu}} \leq \frac{q^{m(n-d+1)}}{q^{mu}} = q^{m(n-d-u+1)}.$$

Definition 3.3. For $u \geq 0$, the u-th normalized binomial moment of C of minimum distance d is defined by

$$b_u(C) := \frac{B_{u+d}(C)}{\begin{bmatrix} n \\ u+d \end{bmatrix}}.$$

Thus, $b_u(C)$ counts the number of non-zero elements of the shortened code C_U where U has dimension $n - u - d$. The definition of b_u is extended to all $u \in \mathbb{Z}$, by

$$b_u(C) = \begin{cases} 0, & \text{if } u < 0. \\ q^{k-m(n-u-d)} - 1, & \text{if } u > n - d^\perp - d. \end{cases}$$

since $u < 0$ implies $B_{u+d} = 0$, and $u > n - d^\perp - d$ implies that $B_{u+d} = q^{k-m(n-u-d)} - 1 \begin{bmatrix} n \\ u+d \end{bmatrix}$.

Remark 3.1. If C is an MRD code, then it has dimension $m(n - d + 1)$ and the dimension of the dual of C is $\dim(C^\perp) = mn - \dim(C)$, which implies that $d^\perp = n - d + 2$. Then $b_u(C) = q^{m(u+1)} - 1$ for all $u \geq 0$. The values of $b_u(C)$ are independent of the minimum distance of C.

Finally, we arrive at the following definition of the zeta function of C for rank-metric codes, given in [3], analogous to Definition 2.6 and Definition 1.11.

Definition 3.4. The *zeta function* of C is defined by

$$Z_C(T) := (q^m - 1)^{-1} \sum_{u \geq 0} b_u(C) T^u.$$

For $u \notin \{0, \ldots, n - d - d^\perp + 2\}$, we have the following relation among the $b_u(C)$

$$b_u(C) - (q^m + 1) b_{u-1}(C) + q^m b_{u-2}(C) = 0. \tag{3.1}$$

For $u < 0$, it is obvious. For $u > n - d - d^\perp$, we have that $b_u(C) = q^{k-m(n-u-d)} - 1 = q^{k-mn+md} \cdot q^{mu} - 1 = A \cdot q^{mu} - 1$,where A is a constant that is independent of u. We will check if (3.1) holds for the generic form $b_u = \alpha q^{mu} + \beta$. Upon substituting $b_u = \alpha q^{mu} + \beta$, $b_{u-1} = \alpha q^{m(u-1)} + \beta$, and $b_{u-2} = \alpha q^{m(u-2)} + \beta$ in (3.1), we get $\alpha q^{mu} + \beta - (q^m + 1)(\alpha q^{-m} q^{mu} + \beta) + q^m(\alpha q^{-2m} q^{mu} + \beta) = 0$.

Definition 3.5. For $u \geq 0$, define

$$p_u(C) := (q^m - 1)^{-1}(b_u(C) - (q^m + 1)b_{u-1}(C) + q^m b_{u-2}(C)). \tag{3.2}$$

The *zeta polynomial* of C is defined by

$$P_C(T) := \sum_{u=0}^{n-d-d^\perp+2} p_u(C)T^u.$$

For any $u > n - d - d^\perp + 2$, $p_u(C) = 0$. We will assume that the minimum distance of the dual code is at least 2, thus,

$$P_C(T) = \sum_{u=0}^{n-d} p_u(C)T^u.$$

Next, we derive the equation that relates the zeta function to the zeta polynomial, analogous to (2.20). Upon multiplying both sides of (3.2) by $(q^m - 1) \cdot T^u$, and summing over $u \geq 0$, we get

$$(q^m - 1)\sum_{u=0}^{\infty} p_u(C)T^u = \sum_{u=0}^{\infty} b_u(C)T^u - (q^m + 1)\sum_{u=0}^{\infty} b_{u-1}(C)T^u + q^m \sum_{u=0}^{\infty} b_{u-2}(C)T^u.$$

The right-hand side simplifies to $(q^m - 1)Z_C(T)[1 - (q^m + 1)T + q^m T^2]$, and the left-hand side is $(q^m - 1)P_C(T)$ by Definition 3.5. Thus we get,

$$Z_C(T) = \frac{P_C(T)}{(1 - T)(1 - q^m T)}. \tag{3.3}$$

This yields the following result for an MRD code, analogous to the zeta polynomial of an MDS code (Proposition 2.1).

Lemma 3.3. Let C be an MRD code. Then $P_C(T) = 1$ and

$$Z_C(T) = \frac{1}{(1 - T)(1 - q^m T)}.$$

Proof. C is MRD if and only if $d + d^\perp = n + 2$. For an MRD code, $b_u = q^{m(u+1)} - 1$ for all $u \geq 0$. Substituting into Definition 3.5 gives $p_0 = 1$ and $p_u = 0$ for all $u \geq 1$. Hence $P_C(T) = 1$. □

3.2 Second and third definition

In this section, we will define the zeta function of a rank-metric code in terms of its weight enumerator polynomial and its MRD expansion of the weight enumerator, analogous to Definition 2.1, and Definition 2.3 in the Hamming metric case. Before doing so, we require the notion of Möbius inversion.

For a partially ordered set P, the Möbius function is defined via the recursive formula

$$\mu(x, x) = 1, \tag{3.4}$$

$$\mu(x, z) = -\sum_{x \leq y < z} \mu(x, y), \text{ for } x < z. \tag{3.5}$$

Lemma 3.4 (Möbius inversion formula)**.** Let $f, g : P \to \mathbb{Z}$ be any two functions on P. Then

1. $f(x) = \sum_{x \leq y} g(y)$ if and only if $g(x) = \sum_{x \leq y} \mu(x, y) f(y)$.

2. $f(x) = \sum_{x \geq y} g(y)$ if and only if $g(x) = \sum_{x \geq y} \mu(x, y) f(y)$.

For the subspace lattice of $\mathbb{F}_q^{m \times n}$ (with partial order defined by set inclusion), and for any two subspaces U and V of dimensions u and v, we have that

$$\mu(U, V) = \begin{cases} (-1)^{v-u} q^{\binom{v-u}{2}}, & \text{if } U \leq V. \\ 0, & \text{otherwise.} \end{cases}$$

Given a subspace U of $\mathbb{F}_q^n$, and using the definition of shortening (Definition 3.1), we have the relation

$$|C_U| = \sum_{U \leq V} |\hat{C}_V|. \tag{3.6}$$

For a fixed U, let $X \in C_U$ be an $X \in C$ such that $U \subseteq X^\perp$. For such an X, let V be a subspace of $\mathbb{F}_q^n$ such that $V = X^\perp$. Then $U \subseteq V$ and $X \in \hat{C}_V$. Conversely, if $X \in \hat{C}_V$ for a fixed V with $U \subseteq V$, then by definition of $\hat{C}_V$, since $V = X^\perp$, we get $U \subseteq X^\perp$, so $X \in C_U$. Thus, $C_U = \sqcup_{U \leq V} \hat{C}_V$, since each X has a unique $X^\perp$. This proves (3.6). And by the Möbius inversion formula, we get

$$|\hat{C}_U| = \sum_{U \leq V} \mu(U, V) |C_V|. \tag{3.7}$$

In the following results, we will derive a more explicit description of $W_C(x, y)$ in terms of the normalized binomial moments.

Lemma 3.5. Let C have minimum distance d and let U be a subspace of $\mathbb{F}_q^n$ of dimension $u < n$. Then

$$|\hat{C}_U| = \sum_{w=0}^{n-d-u} (-1)^w q^{\binom{w}{2}} \sum_{\substack{\dim(V) = u+w \\ U \leq V}} (|C_V| - 1).$$

Proof. Applying (3.4) with $U < \mathbb{F}_q^n$ gives $\mu(U, \mathbb{F}_q^n) = -\sum_{U \leq V < \mathbb{F}_q^n} \mu(U, V) \implies \sum_{U \leq V} \mu(U, V) = 0$, since $U < \mathbb{F}_q^n$. And using Lemma 3.2, we get $|C_V| = 1$, since $\dim(V) = v > n - d$. Thus, from (3.7),

$$\begin{aligned} |\hat{C}_U| &= \sum_{U \leq V} \mu(U, V) |C_V| = \sum_{U \leq V} \mu(U, V)(|C_V| - 1) + \sum_{U \leq V} \mu(U, V) \\ &= \sum_{U \leq V} \mu(U, V)(|C_V| - 1) = \sum_{\substack{U \leq V \\ v \leq n-d}} \mu(U, V)(|C_V| - 1) = \sum_{\substack{U \leq V \\ v \leq n-d}} (-1)^{v-u} q^{\binom{v-u}{2}} (|C_V| - 1) \\ &= \sum_{w=0}^{n-d-u} (-1)^w q^{\binom{w}{2}} \sum_{\substack{\dim(V) = u+w \\ U \leq V}} (|C_V| - 1). \end{aligned}$$

□

Proposition 3.1. *Let C have minimum distance d. Then*

$$W_t(C) = \sum_{i=d}^{t} (-1)^{t-i} q^{\binom{t-i}{2}} \begin{bmatrix} n \\ t \end{bmatrix} \begin{bmatrix} t \\ i \end{bmatrix} b_{i-d}(C). \tag{3.8}$$

Proof. We recall the definition $W_t(C) = |\{X \in C | \operatorname{rk}(X) = t\}|$. For any $X \in \mathbb{F}_q^{m\times n}$, we had that $\operatorname{rk}(X) = n - \dim(X^\perp)$. Thus, $\operatorname{rk}(X) = t \iff \dim(X^\perp) = n - t$. Let $u = n - t$, then $\operatorname{rk}(X) = t \iff \dim(X^\perp) = u$. Thus, $W_{n-u}(C)$ counts matrices whose nullspace has dimension u. Since every matrix $X \in C$ has a unique nullspace, we have $\{X \in C : \dim(X^\perp) = u\} = \sqcup_{\dim(U)=u}\{X \in C : X^\perp = U\} = \sqcup_{\dim(U)=u}\hat{C}_U$. Thus, $W_{n-u}(C) = \sum_{\dim(U)=u} |\hat{C}_U|$. Applying Lemma 3.5 and setting $u + w = v$ gives

$$\sum_{w=0}^{n-d-u} (-1)^w q^{\binom{w}{2}} \sum_{U\le V} (|C_V| - 1).$$

For a fixed V, there are $\binom{u+w}{u}$U's contained in V of dimension u. Hence

$$W_t(C) = \sum_{w=0}^{n-d-u} (-1)^w q^{\binom{w}{2}} \begin{bmatrix} u+w \\ u \end{bmatrix} B_{n-(u+w)}(C)$$

which equals

$$\sum_{w=0}^{t-d}(-1)^w q^{\binom{w}{2}} \begin{bmatrix} n \\ n-t+w \end{bmatrix} \begin{bmatrix} n-t+w \\ n-t \end{bmatrix} b_{t-w-d}(C) = \sum_{w=0}^{t-d}(-1)^w q^{\binom{w}{2}} \begin{bmatrix} n \\ t \end{bmatrix} \begin{bmatrix} t \\ w \end{bmatrix} b_{t-w-d}(C).$$

Let $t - w = i$ and using $\begin{bmatrix} t \\ w \end{bmatrix} = \begin{bmatrix} t \\ t-w \end{bmatrix}$, we get (3.8). □

Corollary 3.1. *Let C have minimum distance d. Then*

$$W_C(x,y) = x^n + \sum_{i=d}^{n}\left(\sum_{t=i}^{n}(-1)^{t-i}q^{\binom{t-i}{2}} \begin{bmatrix} n \\ t \end{bmatrix} \begin{bmatrix} t \\ i \end{bmatrix} x^{n-t}y^t\right) b_{i-d}(C). \tag{3.9}$$

In particular, in the case of an MRD code with weight enumerator $M_{n,d}(x,y)$, we get

$$M_{n,d}(x,y) - x^n = \sum_{i=d}^{n}\left(\sum_{t=i}^{n}(-1)^{t-i}q^{\binom{t-i}{2}} \begin{bmatrix} n \\ t \end{bmatrix} \begin{bmatrix} t \\ i \end{bmatrix} x^{n-t}y^t\right)(q^{m(i-d+1)} - 1). \tag{3.10}$$

Proof. (3.9) follows by substituting (3.8) in (1.4). And since $b_{i-d}(C) = q^{m(i-d+1)} - 1$, (3.10) follows. □

Definition 3.6. For each $r \in \{0,\dots,n\}$ define

$$\phi_{n,r}(x,y) := (q^m - 1)^{-1}\left(M_{n,r}(x,y) - (q^m+1)M_{n,r+1}(x,y) + q^m M_{n,r+2}(x,y)\right),$$

and

$$\phi_n(T) := \sum_{r=0}^{n}\phi_{n,r}(x,y)T^r.$$

Therefore,

$$\frac{\phi_n(T)}{(1-T)(1-q^mT)} \equiv (q^m-1)^{-1}\sum_{r=0}^{n}\left(M_{n,r}(x,y) - x^n\right)T^{n-r} \pmod{T^{n+1}}. \tag{3.11}$$

Lemma 3.6. For each $r \in \{0,\dots,d\}$, the coefficient of T^{n-r} in the expression,

$$Z_C(T)\phi_n(T) = \frac{P_C(T)\phi_n(T)}{(1-T)(1-q^mT)}$$

is given by

$$(q^m-1)^{-1}\left(\sum_{i=0}^{n-r} p_i(C)M_{n,r+i}(x,y)-x^n\right). \tag{3.12}$$

Proof. From (3.11) and (3.3), we get that

$$Z_C(T)\cdot\phi_n(T) = \frac{P_C(T)}{(1-T)(1-q^mT)}\phi_n(T)$$

$$= P_C(T)((q^m-1)^{-1}\sum_{s=0}^{n}(M_{n,s}(x,y)-x^n)T^{n-s} + \text{higher order terms.} \tag{3.13}$$

We know that $P_C(T) = \sum_{i=0}^{n-d} p_i(C)T^i$ has degree at most $n-d$. In the product on the right-hand side of (3.13), the exponent from $P_C(T)$ is i, and from the series it is $n-s$. To obtain $n-r$, we require that $i+(n-s)=n-r \implies s=r+i$. Thus, the coefficient of T^{n-r} is

$$\frac{1}{(q^m-1)}\left(\sum_{i=0}^{n-r} p_i(C)M_{n,r+i}(x,y)-x^n\sum_{i=0}^{n-r}p_i(C)\right). \tag{3.14}$$

From Definition 3.5, we have the definition of $p_i(C)$, and we can evaluate $\sum_{i=0}^{n-r} p_i(C)$ to be

$$(q^m-1)\sum_{i=0}^{n-r}p_i(C) = \sum_{i=0}^{n-r}\left(b_i(C)-(q^m+1)b_{i-1}(C)+q^mb_{i-2}(C)\right). \tag{3.15}$$

The sums on the right-hand side of (3.15) expand in the following way:

$$\sum_{i=0}^{n-r} b_i(C) = b_0(C)+b_1(C)+\cdots+b_{n-r}(C) \tag{3.16}$$

$$-(q^m+1)\sum_{i=0}^{n-r} b_{i-1}(C) = -(q^m+1)(b_{-1}(C)+b_0(C)+\cdots+b_{n-r-1}(C)) \tag{3.17}$$

$$q^m\sum_{i=0}^{n-r} b_{i-2}(C) = q^m\left(b_{-2}(C)+b_{-1}(C)+b_0(C)+\cdots+b_{n-r-2}(C)\right) \tag{3.18}$$

Upon adding (3.16), (3.17), and (3.18), and using the fact that $b_u=0$ for $u<0$, we get the remaining terms to be

$$b_{n-r}(C)-q^mb_{n-r-1}(C)-b_{-1}(C)+q^mb_{-2}(C) = b_{n-r}(C)-q^mb_{n-r-1}(C).$$

Thus we get

$$\sum_{i=0}^{n-r}p_i(C) = \frac{b_{n-r}(C)-q^mb_{n-r-1}(C)}{q^m-1}. \tag{3.19}$$

Since $r\le d$, we have $n-r\ge n-d$. And since $d^\perp\ge 2$, it follows that $n-r-1\ge n-d-1>n-d-d^\perp$. Hence $n-r$ and $n-r-1$ are both greater than $n-d-d^\perp$, so by Definition 3.3 we have $b_{n-r}(C)=q^{k-m(r-d)}-1$ and $b_{n-r-1}(C)=q^{k-m(r+1-d)}-1$. Substituting these in (3.19) gives us $\sum_{i=0}^{n-r}p_i(C)=1$, which we substitute in (3.14) to get the coefficient of T^{n-r} to be (3.12). □

An explicit expression for $\phi_n(T)$ is given in the following way.

Lemma 3.7.

$$\phi_n(T) = \sum_{i=0}^{n} \left(\sum_{t=i}^{n} (-1)^{t-i} q^{\binom{t-i}{2}} \begin{bmatrix} n \\ t \end{bmatrix} \begin{bmatrix} t \\ i \end{bmatrix} x^t y^{n-t} \right) T^{n-i}.$$

Proof. By Definition 3.6, we have that

$$(q^m - 1)\phi_{n,n-r}(x,y) = M_{n,r}(x,y) - (q^m+1)M_{n,r+1}(x,y) + q^m M_{n,r+2}(x,y).$$

Upon substituting the values of $M_{n,r}$, $M_{n,r+1}$, and $M_{n,r+2}$ from (3.10), we get

$$\begin{aligned}
&= \sum_{t=r}^{n} (-1)^{t-r} q^{\binom{t-r}{2}} \begin{bmatrix} n \\ t \end{bmatrix} \begin{bmatrix} t \\ r \end{bmatrix} x^t y^{n-t} (q^m - 1) \\
&\quad + \sum_{t=r+1}^{n} (-1)^{t-r-1} q^{\binom{t-r-1}{2}} \begin{bmatrix} n \\ t \end{bmatrix} \begin{bmatrix} t \\ r+1 \end{bmatrix} x^t y^{n-t} (q^{2m} - 1 - (q^m+1)(q^m-1)) \\
&\quad + \sum_{i=r+2}^{n} \sum_{t=i}^{n} (-1)^{t-i} q^{\binom{t-i}{2}} \begin{bmatrix} n \\ t \end{bmatrix} \begin{bmatrix} t \\ i \end{bmatrix} x^t y^{n-t} (b_{i-r+1} - (q^m+1)b_{i-r} + q^m b_{i-r-1}) \\
&= \sum_{t=r}^{n} (-1)^{t-r} q^{\binom{t-r}{2}} \begin{bmatrix} n \\ t \end{bmatrix} \begin{bmatrix} t \\ r \end{bmatrix} x^t y^{n-t} (q^m - 1).
\end{aligned}$$

□

Thus, the weight enumerators $M_{n,d}(x,y)$ and $W_C(x,y)$ can be expressed as

$$M_{n,d}(x,y) = x^n + \sum_{i=d}^{n} \phi_{n,n-i}(x,y) b_{i-d}. \tag{3.20}$$

$$W_C(x,y) = x^n + \sum_{i=d}^{n} \phi_{n,n-i}(x,y) b_{i-d}(C). \tag{3.21}$$

The polynomials $\phi_{n,n-r}(x,y)$ are related to a class of $q-$polynomials [21], and are given by

$$\phi_{n,n-r}(x,y) = \begin{bmatrix} n \\ r \end{bmatrix} p_{n-r}(x,y) y^r,$$

where $p_k(x,y) := \prod_{j=0}^{k-1}(x - q^j y) = \sum_{j=0}^{k} (-1)^{k-j} q^{\binom{k-j}{2}} \begin{bmatrix} k \\ j \end{bmatrix} x^j y^{k-j}$. Therefore,

$$W_C(x,y) = x^n + \sum_{i=d}^{n} b_{i-d}(C) \begin{bmatrix} n \\ i \end{bmatrix} p_{n-i}(x,y) y^i. \tag{3.22}$$

Next, we establish a connection between the weight enumerator of a rank-metric code and its zeta function, from which we derive two definitions of the zeta function of rank-metric codes, analogous to Definition 2.1 and Definition 2.3 of the zeta function of the Hamming metric code.

Theorem 3.1. *The coefficient of T^{n-d} in the expression,*

$$Z_C(T)\phi_n(T) = \frac{P_C(T)\phi_n(T)}{(1-T)(1-q^m T)},$$

is given by

$$\frac{W_C(x,y) - x^n}{q^m - 1}.$$

In particular, the zeta polynomial $P_C(T) = \sum_{i\geq 0} p_i T^i$ *is the unique polynomial of degree at most* $n-d$ *such that* $W_C(x,y) = \sum_{i=0}^{n-d} p_i M_{n,d+i}(x,y)$.

Proof. We have from (3.22) that the coefficient of T^{n-d} in $Z_C(T)\phi_n(T)$ is $(q^m-1)^{-1}(W_C(x,y) - x^n)$. From Lemma 3.6, we have that this coefficient is equal to $(q^m-1)^{-1}(-x^n + \sum_{i=0}^{n-d} p_i M_{n,d+i}(x,y))$, and thus $W_C(x,y) = \sum_{i=0}^{n-d} p_i(C) M_{n,d+i}(x,y)$. Since the $M_{n,d+i}(x,y)$ are linearly independent over $\mathbb{Q}$, we get the required result. □

Next, we derive the duality condition for the zeta functions of rank-metric codes, analogous to (1.12) for algebraic curves and (2.4) for Hamming metric codes. To do so, we require the duality relation of MRD codes. Since an MRD code exists for all choices of m, n, d, the family of MRD weight enumerators is closed under the Mac Williams identity. We explicitly have the following expression for the dual of an MRD code.

Theorem 3.2 ([3], Theorem 2)**.** $M^{\perp}_{n,d}(x,y) = q^{m(n-d+1)} M_{n,n-d+2}(x,y)$.

From Theorem 3.3 ([16]), we know that the Mac Williams transform relating the weight enumerator of a code with that of its dual is a $\mathbb{Q}-$linear map. We will apply this fact along with Theorem 3.2 to get the following corollary.

Corollary 3.2. *Let* $C \subseteq \mathbb{F}_q^{m\times n}$ *be a matrix code with minimal rank distance* d *and dimension* k*. Let* $d^{\perp}$ *be the minimal rank distance of the dual code* $C^{\perp}$*. Setting*

$$W_C(x,y) = \sum_{i=0}^{r} p_i(C) M_{n,d+i} \quad \text{and} \quad W_{C^{\perp}}(x,y) = \sum_{j=0}^{t} s_j(C^{\perp}) M_{n,d^{\perp}+j},$$

and letting $p_i(C) = p_i$*, and* $s_j(C^{\perp}) = s_j$ *we have that*

1. $r = t = n - d - d^{\perp} + 2$,
2. $s_j = p_{r-j}\, q^{m(d^{\perp}+j-1)-k}$.

Proof. Since the transform in Theorem 3.2 is $\mathbb{Q}-$linear, we can apply Mac Williams transform term wise to $W_C(x,y)$ to get

$$W_{C^{\perp}}(x,y) = \sum_{i=0}^{n-d} p_i (M_{n,d+i})^{\perp}.$$

$$W_{C^{\perp}}(x,y) = \sum_{i=0}^{n-d} p_i q^{m(n-d-i+1)} M_{n,n-d-i+2}(x,y). \tag{3.23}$$

And by using the Mac Williams transform (Theorem 3.3, [16]), we get that $q^k W_{C^{\perp}}$ is equal to the right-hand side of (3.23), which gives us

$$W_{C^{\perp}}(x,y) = \sum_{i=0}^{r} q^{m(n-d-i+1)-k} p_i M_{n,n-d-i+2}. \tag{3.24}$$

Setting $d^{\perp} + j = n - d - i + 2$, we have $j = 0$ when $i = n - d - d^{\perp} + 2$, and correspondingly we get the maximal value of j when $i = 0$, i.e., $j = n - d - d^{\perp} + 2$. Thus, (1) follows. To prove (2), we need to

write (3.24) in the form of $W_{C^\perp}(x,y) = \sum_{j=0}^{t} s_j(C^\perp) M_{n,d^\perp+j}$. This follows because the MRD family is a $\mathbb{Q}-$basis, and hence the decomposition of $W_{C^\perp}$ as a linear combination of MRD polynomials of different minimal distances must be unique. Let's set $d^\perp + j = n - d - i + 2$, which implies that $i = r - j$ (from (1)). Upon substituting this into (3.24), the exponent of q simplifies to $m(d^\perp + j - 1) - k$, giving us (2). □

The Mac Williams identity can be translated into a functional relation between the zeta function of a code and that of its dual.

Theorem 3.3 (Functional equation of the zeta function of a rank-metric code)**.**

$$Z_{C^\perp}(T) = q^{m(n-d+1)+k} T^{r-2} Z_C\left(\frac{1}{q^m T}\right). \tag{3.25}$$

In particular, if C is formally self-dual, then

$$Z_C(T) = q^{m(n-d+1)+k} T^{r-2} Z_C\left(\frac{1}{q^m T}\right). \tag{3.26}$$

Proof. From Corollary 3.2, we know $W_C(x,y) = \sum_{i=0}^{r} p_i M_{n,d+i}$ and $W_{C^\perp}(x,y) = \sum_{j=0}^{t} s_j M_{n,d^\perp+j}$. Thus, we have $P_C(T) = \sum_{i=0}^{r} p_i T^i$ and $P_{C^\perp}(T) = \sum_{j=0}^{r} s_j T^j$, for some $p_i, s_j \in \mathbb{Q}$. From Corollary 3.2, we have

$$s_j = p_{r-j}\, q^{m(d^\perp+j-1)-k}. \tag{3.27}$$

Upon multiplying both sides of (3.27) by T^{j-r} and summing in j, and then changing the index from j to $r - j$, we get

$$T^{-r} P_{C^\perp}(T) = \sum_{j=0}^{r} p_j q^{m(d^\perp+r-j-1)-k} T^{-j} = q^{m(d^\perp+r-1)-k} P_C\left(\frac{1}{q^m T}\right).$$

Therefore, we get

$$P_{C^\perp}(T) = T^r q^{m(n-d+1)-k} P_C\left(\frac{1}{q^m T}\right), \tag{3.28}$$

which gives us (3.25) after we quotient $(1-T)(1-q^m T)$ on both sides. □

Remark 3.2. In Example 2 of [3], the authors give two examples of codes that have the same zeta polynomials as their duals, but not the same weight enumerators. This is due to the fact that the minimum rank distances d and $d^\perp$ differ. Conversely, if two codes have the same weight enumerator, then they have the same zeta polynomial, and hence the same zeta function. This is because the zeta polynomial is uniquely determined by the weight enumerator via Theorem 3.1. This is analogous to the case of the Hamming metric, as discussed in Remark 2.2.

In fact, if a code has a quadratic zeta polynomial, which occurs if and only if $d + d^\perp = n$ since $\deg P_C = n - d - d^\perp + 2$, then its dual will have the same zeta polynomial if and only if $k = m(n-d)$. We can explicitly derive this property from the functional equation of the zeta polynomial given by Theorem 3.3. For a quadratic zeta polynomial (3.28) becomes

$$P_{C^\perp}(T) = T^2 q^{m(n-d+1)-k} P_C(1/q^m T),$$

and since we want the dual to have the same zeta polynomial as the original code, this becomes

$$P_C(T) = T^2 q^{m(n-d+1)-k} P_C(1/q^m T).$$

Let $P_C(T)$ be the quadratic polynomial given by

$$P_C(T) = p_0 + p_1T^1 + p_2T^2 \tag{3.29}$$

(with $p_2 \neq 0$). Then

$$P_C(1/q^mT) = p_0 + p_1(q^mT)^{-1}T + p_2(q^mT)^{-2}.$$

Upon multiplying on both sides by T^2 and $q^{m(n-d+1)-k}$, we get

$$q^{m(n-d+1)-k}T^2P_C(\frac{1}{q^mT}) = q^{m(n-d+1)-k}p_2(q^m)^{-2}+q^{m(n-d+1)-k}p_1(q^m)^{-1}T+q^{m(n-d+1)-k}p_0T^2. \tag{3.30}$$

Upon equating the coefficient of T^1 in (3.29) and (3.30), we get

$$p_1 = q^{m(n-d+1)-k-m}p_1$$

which gives us $m(n-d+1)-k-m = 0 \implies k = m(n-d)$.

3.3 The Riemann hypothesis analogue: an open problem

Analogous to the reciprocal zeros of the zeta polynomial of an algebraic curve (Corollary 1.3) and Hamming metric codes (Proposition 2.6), the reciprocal zeros of rank-metric codes are important. They are used to obtain upper bounds on their minimum rank distance. We explicitly show this relation in the following theorem.

Theorem 3.4. *Let $P_C(T) = p_0(1 + a'T + \cdots)$ be the zeta polynomial of a rank-metric code $C \subseteq \mathbb{F}_q^{m\times n}$, of degree $r \leq n-d$ and let a' be the negative of the sum of its reciprocal zeros. Then*

$$d \leq \log_q\left[(a' + q^m + 1)(q-1) + 1\right] - 1. \tag{3.31}$$

Proof. First, we observe that according to Theorem 2.1, the coefficient of T^{n-d} in $Z_C(T)\phi_n(T)$ is

$$\frac{W_C(x,y) - x^n}{q^m - 1}. \tag{3.32}$$

Second, since $\frac{1}{(1-T)(1-q^mT)}$ is the common zeta function of all the MRD weight enumerators (in particular, those with length n), we get from Theorem 2.1 that

$$\frac{\phi_n(T)}{(1-T)(1-q^mT)} = \frac{1}{q^m-1}\sum_{i=0}^{n-d}(M_{n,d+i}(x,y) - x^n)T^{n-d-i} + \text{ (higher order terms in T) .}$$

And similarly we get the coefficient of T^{n-d} in $Z_C(T)\phi_n(T)$ is

$$\frac{1}{q^m-1}\sum_{i=0}^{r}p_i(M_{n,d+i}(x,y) - x^n). \tag{3.33}$$

From the two coefficients of T^{n-d} that we got from (3.32) and (3.33), we want to compare the coefficients of $x^{n-d}y^d$ and $x^{n-d-1}y^{d+1}$. The coefficient of $x^{n-d}y^d$ in (3.32) is $\frac{W_d(C)}{q^m-1}$, and this equals the coefficient of $x^{n-d}y^d$ in (3.33), i.e., $\frac{p_0}{q^m-1}(M_{n,d}(x,y) - x^n)$. From (1.5), we get the coefficient of $x^{n-d}y^d$ to be

$\begin{bmatrix} n \\ d \end{bmatrix}(q^m-1) = \begin{bmatrix} n \\ d \end{bmatrix} b_0$. Thus the coefficient of $x^{n-d}y^d$ in (3.33) is $\frac{p_0}{q^m-1}\begin{bmatrix} n \\ d \end{bmatrix} b_0$. Upon equating both the coefficients, we get that $p_0 = \frac{W_d(C)}{(q^m-1)\begin{bmatrix} n \\ d \end{bmatrix}} \geq 0$. Next, we compare the coefficients of $x^{n-d-1}y^{d+1}$. From (3.32), it is $\frac{W_{d+1}(C)}{q^m-1} \geq 0$. And it equals the coefficient of $x^{n-d-1}y^{d+1}$ from (3.33) from

$$\frac{1}{q^m-1}\left(p_0\left(M_{n,d}(x,y)-x^n\right)+p_1\left(M_{n,d+1}(x,y)-x^n\right)\right).$$

The coefficient in $\frac{p_1}{q^m-1}\left(M_{n,d+1}(x,y)-x^n\right)$ is like before

$$\frac{p_1}{q^m-1}\begin{bmatrix} n \\ d+1 \end{bmatrix} b_0. \tag{3.34}$$

As for the coefficient in $\frac{p_0}{q^m-1}\left(M_{n,d}(x,y)-x^n\right)$, we expand the expression using (3.10) as

$$\frac{p_0}{q^m-1}\left(\sum_{i=d}^{n}\left(\sum_{t=i}^{n}(-1)^{t-i}q^{\binom{t-i}{2}}\begin{bmatrix} n \\ t \end{bmatrix}\begin{bmatrix} t \\ i \end{bmatrix}x^{n-t}y^t\right)\left(q^{m(i-d+1)}-1\right)\right).$$

Let $t = n-j$, then $j = n-t$ and we get

$$\begin{aligned}&\frac{p_0}{q^m-1}\left(\sum_{i=d}^{n}\left(\sum_{j=0}^{n}(-1)^{n-j-i}q^{\binom{n-i-j}{2}}\begin{bmatrix} n \\ i \end{bmatrix}\begin{bmatrix} n-i \\ j \end{bmatrix}x^j y^{n-j}\right)\left(q^{m(i-d+1)}-1\right)\right)\\ &=\sum_{i=d}^{n}\begin{bmatrix} n \\ i \end{bmatrix}b_{i-d}y^i\sum_{j=0}^{n-i}(-1)^{n-i-j}q^{\binom{n-i-j}{2}}\begin{bmatrix} n-i \\ j \end{bmatrix}x^j y^{n-i-j},\end{aligned}$$

from which we get the coefficient of $x^{n-d-1}y^{d+1}$ to be

$$\frac{p_0}{q^m-1}\left(\begin{bmatrix} n \\ d+1 \end{bmatrix}b_1-\begin{bmatrix} n \\ d \end{bmatrix}\begin{bmatrix} n-d \\ n-d-1 \end{bmatrix}b_0\right). \tag{3.35}$$

Adding (3.34) and (3.35), dividing by p_0, keeping in mind that $p_1/p_0 = a'$, and using $\begin{bmatrix} n \\ d+1 \end{bmatrix} = \frac{q^{n-d}-1}{q^{d+1}-1}\begin{bmatrix} n \\ d \end{bmatrix}$, gives

$$a'b_0\frac{q^{n-d}-1}{q^{d+1}-1}+b_1\frac{q^{n-d}-1}{q^{d+1}-1}-b_0\begin{bmatrix} n-d \\ n-d-1 \end{bmatrix}\geq 0. \tag{3.36}$$

And since $\begin{bmatrix} n-d \\ n-d-1 \end{bmatrix} = \frac{q^{n-d}-1}{q-1}$, it holds that,

$$a'b_0+b_1-b_0\frac{q^{d+1}-1}{q-1}\geq 0.$$

Since $b_u = q^{m(u+1)} - 1$ for each $u \geq 0$, dividing by b_0 gives us

$$a' + q^m + 1 \geq \frac{q^{d+1} - 1}{q - 1}.$$

Taking logarithm on both sides yields (3.31). □

The next natural question that one would ask is whether there exists a Riemann hypothesis analogue for the rank-metric codes. From Theorem 3.3, we can obtain the functional equation of the zeta polynomial of a rank-metric code to be

$$P_{C^\perp}(T) = q^{m(n-d+1)-k} T^r P_C\left(\frac{1}{q^m T}\right). \tag{3.37}$$

Suppose C is formally self-dual, and α is a root of $P_C(T) = 0$, where $\alpha \neq 0$, then from (3.37), we get that $P_C(1/(q^m\alpha)) = 0$. Thus, the non-zero (in general complex) zeros of the zeta polynomial $P_C(T)$ occur in pairs $(\alpha, 1/(q^m\alpha))$. The two zeros in the pair are of the same absolute value if

$$|\alpha| = \left|\frac{1}{q^m\alpha}\right| \implies |\alpha|^2 = q^{-m} \implies |\alpha| = q^{\frac{-m}{2}}.$$

Upon changing the variable from T to $(q^m)^{-s}$, we get $\zeta_C(s) = Z_C((q^m)^{-s})$, and we substitute it in the condition $|\alpha| = |(q^m)^{-s}| = q^{\frac{-m}{2}}$, which gives $(q^m)^{-\mathrm{Re}(s)} = q^{\frac{-m}{2}}$. Thus the zeros of $\zeta_C(s)$ must satisfy $\mathrm{Re}(s) = 1/2$. This is the critical line for the classical Riemann zeta function (Section 1.1), as well as for the zeta function of an algebraic curve over a finite field (Subsection 1.4.3).

Despite promising leads for the existence of a Riemann hypothesis analogue for rank-metric codes, we do not have one yet, not even for some specific families of rank-metric codes.

The authors of [3] give some specific examples of rank-metric codes, that satisfy the Riemann hypothesis analogue. In Example 6 of [3], the authors construct a 4×4 rank-metric code over $\mathbb{F}_2$, by applying certain puncturing and shortening operations on the extended binary QR-code of length 18 in double circulant form. Its zeta polynomial is given by $P_C(T) = (1 + 4T)^2/25$, which has a double zero at $T = -1/4$. The double zero has absolute value $(q^m)^{-1/2} = (2^4)^{-1/2} = 1/4$ and thus the zeros of the zeta function lie on the critical line. The zeta function is that of a maximal elliptic curve over the field of 16 elements.

They remark (Remark 2, [3]) that constructing a self-dual rank-metric code from a self-dual linear code in this way is not unique; in other words, equivalent codes of length 16 may lead to inequivalent 4×4 rank-metric codes. The rank weight distributions obtained need not bear any relations to each other. It requires further research as to what distributions, and hence what zeta polynomials, may occur from codes constructed in this way. They applied a similar construction to another example, to double circulant codes of length 38 to obtain formally self-dual 6×6 rank metric codes. Among those, 12 codes had minimum rank distance 3 and thus a quadratic zeta polynomial (since $\deg P_C = n - d - d^\perp + 2 = 6 - 3 - 3 + 2 = 2$). The rank distributions for these codes are not far apart and in all 12 cases the zeros of the zeta polynomial are on the critical line. Among codes with lower minimum rank distance a wider range of rank distributions occurs. The distributions close to the average have zeros on the critical line and those farther away have pairs of real zeros. Typically one of the real zeros among $(\alpha, 1/(q^m\alpha))$ is close to 1 and the other very close to $1/q^m$. This agrees with the observations for the Hamming metric, where real zeros (when they occur) appear in pairs near 1 and $1/q$.

Despite differences in the metrics, structural consequences of the functional equation lead to similar zero distributions in both settings when the Riemann hypothesis analogue fails. These preliminary observations

suggest that the zeros of the zeta polynomials for the rank-metric have properties similar to those for the Hamming metric. It remains to make these properties precise and to establish to what extent or under which additional assumptions the properties are similar to those for curves.

The authors of [3] also give an example of a family of rank-metric codes, that do not satisfy the Riemann hypothesis analogue in Section 7.1 of [3]. They consider a family of codes which are divisible (that is, the rank weight of every codeword is divisible by some integer $c > 1$) and also formally self-dual. We know that for the Hamming metric, such codes can only exist for $(q, c) \in \{(2, 2), (2, 4), (3, 3), (4, 2)\}$ (Theorem 1.1). In [2], as mentioned in Subsection 2.4, the zeta polynomials of self-dual divisible codes that are also *random* were considered. The authors of [3] show the existence of formally self-dual divisible codes for $c = 2$ and all values of q. However, they note that this construction does not lead to random codes. They also observe that its zeta polynomial has two distinct real roots whose absolute values are not equal to $q^{-m/2}$, proving an explicit counterexample to the Riemann hypothesis analogue.

Conclusion

In this thesis, we explored the theory of zeta functions of linear codes in the Hamming and rank-metrics, based primarily on the work of Duursma and of Blanco-Chacón, Byrne, Duursma, and Sheekey. We studied the structure of the zeta functions of Hamming and rank-metric codes, their properties, and their functional equations. We discussed the Riemann hypothesis analogues for linear codes, which are inspired by the Weil conjectures for algebraic curves over finite fields, and highlighted the structural analogies between them.

For Hamming metric codes, we focused on self-dual codes due to their stronger analogy with algebraic curves. The Riemann hypothesis analogue holds for extremal Type IV self-dual codes, as proved by Duursma. For Types I, II, and III, the question remains open. For rank-metric codes, the situation is more complicated. Unlike the Hamming metric case, where the Riemann hypothesis analogue holds for extremal Type IV codes, the rank-metric codes admit explicit counterexamples. While some families satisfy the critical line condition, a general Riemann hypothesis analogue for rank-metric codes remains an open problem.

There are several directions for future work. The Riemann Hypothesis analogue for extremal self-dual Hamming codes of Types I, II, and III can be explored further. We can characterize which families of rank-metric codes satisfy the critical line condition. And there is also the possibility to extend the theory to sum-rank metric codes.

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